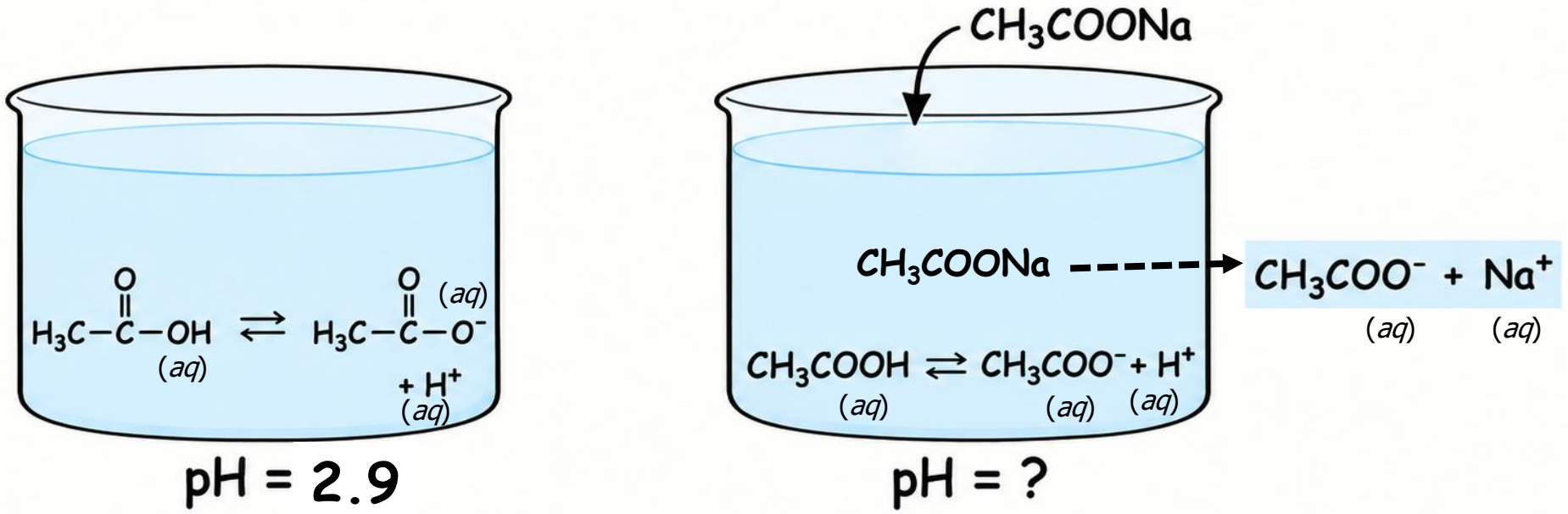
Chapter 17

Additional Aspects of Aqueous Equilibria

- The Common-Ion Effect
- Buffers
- Acid–Base Titrations
- Solubility Equilibria
- Factors That Affect Solubility
- Precipitation and Separation of Ions

The Common-Ion Effect

Q. A solution of acetic acid has pH 2.9. Solid sodium acetate is dissolved in it. Will the pH go up, go down, or stay the same? Explain with Le Châtelier's principle.



CH_3COO^- is a product of the acetic-acid equilibrium, so when more CH_3COO^- is added, the equilibrium shifts left, $[H^+]$ decreases and the pH goes **up**.

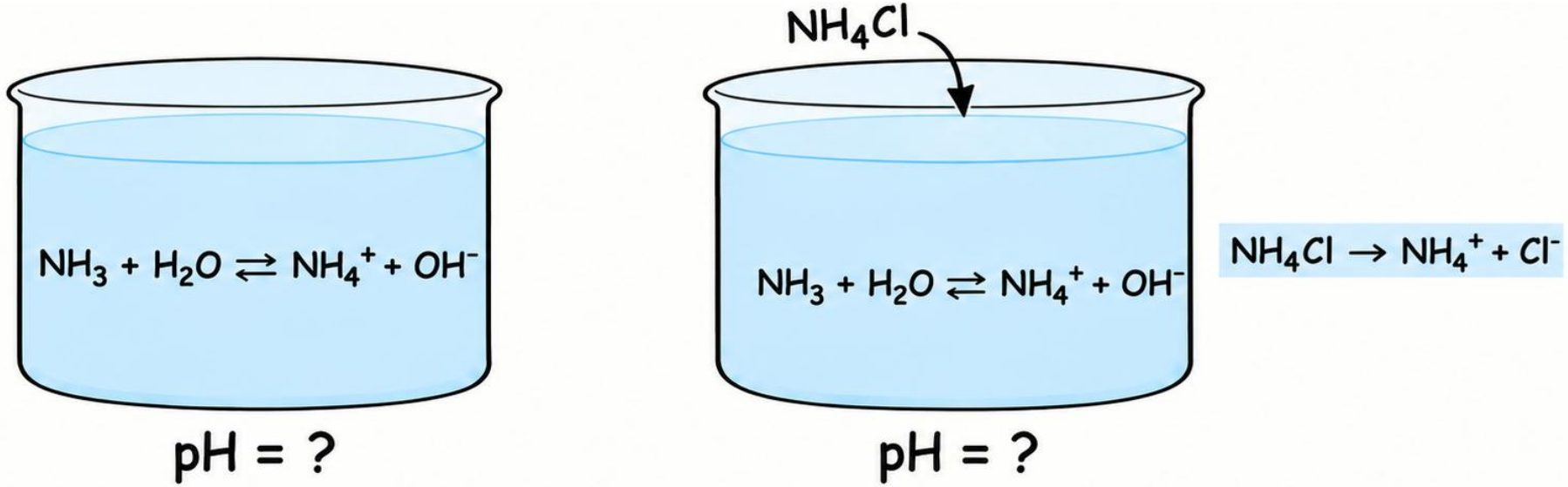
$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

Addition of CH_3COO^- shifts equilibrium concentrations, lowering $[H^+]$

- This kind of observation is called the **common-ion effect**.

The Common-Ion Effect

Common-ion effects for weak bases work the same way



Q. Compare the pH of the $NH_3(aq)$ solution before and after adding NH_4Cl (aq).

$$NH_3(aq) + H_2O\ (l) \rightleftharpoons NH_4^+\ (aq) + OH^-\ (aq)$$

Addition of NH_4^+ shifts equilibrium concentrations, lowering $[OH^-]$

The Common-Ion Effect

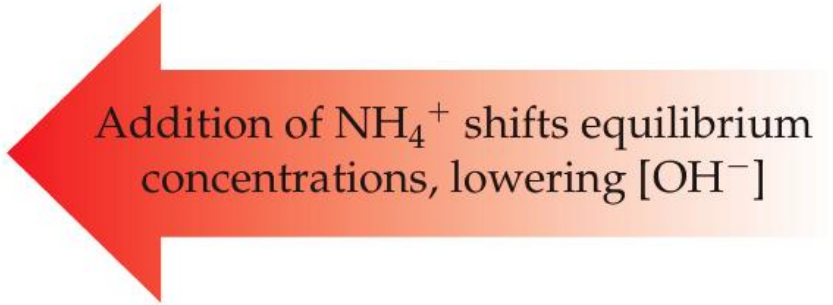
The Common-Ion Effect

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$



*"Whenever a weak electrolyte and a strong electrolyte containing a common ion are together in solution, the **weak electrolyte ionizes less than it would if it were alone in solution**."*

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$



- This affects acid–base equilibria and solubility.

Q. Will the equilibrium constant change due to common ion effect?

The Common-Ion Effect

Q. What is the pH of a solution made by adding 0.30 mol of acetic acid and 0.30 mol of sodium acetate to enough water to make 1.0 L of solution? (K_a = 1.8 × 10^{-5})

1. $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$

2. $K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]}$ = 1.8 × 10^{-5}

3. ICE Table

	$[CH_3COOH]$ (M)	$[H^+]$ (M)	$[CH_3COO^-]$ (M)
I (M)	0.30	0	**0.30**
C (M)	−x	+x	+x
E (M)	0.30 − x	x	0.30 + x

4. $K_a = \frac{(x)(0.30+x)}{(0.30-x)}$ = 1.8 × 10^{-5}

$K_a = \frac{(x)(0.30)}{(0.30)}$ = 1.8 × 10^{-5}

x = 1.8 × 10^{-5} M = $[H^+]$

pH = −log$[H^+]$ = **4.74**

Q. Without the salt, the pH of acetic acid (*aq*) would be:

a. < 4.74 **b.** > 4.74 **c.** = 4.74

The Common-Ion Effect

Q. Calculate the fluoride ion concentration and pH of a solution that is 0.20 M in HF and 0.10 M in HCl. (*Ans.* pH = 1.0)

Q. Calculate the concentration of the lactate ion in a solution that is 0.100 M in lactic acid ($CH_3CH(OH)COOH$, pKa = 3.86) and 0.080 M in HCl.

Q. Calculate the pH of a solution that is 0.250 M in sodium methanoate (HCOONa) and 0.150 M in methanoic acid (HCOOH, $K_a = 1.8 \times 10^{-4}$).

Q. Calculate the pH of a solution that is made by combining 80 mL of 0.050 M hydrofluoric acid ($K_a = 6.8 \times 10^{-4}$) with 100 mL of 0.10 M sodium fluoride.

Q. Calculate the pH of a solution that is 0.490 M in pyridine (C_5H_5N, $K_b = 1.7 \times 10^{-9}$) and 0.450 M in pyridinium chloride (C_5H_5NHCl).

The Common-Ion Effect

Q. Calculate the pH of a solution that is 0.490 M in pyridine (C_5H_5N, K_b = 1.7 × 10^{-9}) and 0.450 M in pyridinium chloride (C_5H_5NHCl).

1. $C_5H_5N(aq) + H_2O(l) \rightleftharpoons C_5H_5NH^+(aq) + OH^-(aq)$

2. $K_b = \frac{[C_5H_5NH^+][OH^-]}{[C_5H_5N]} = 1.7 \times 10^{-9}$

3. ICE table

	C_5H_5N (M)	$C_5H_5NH^+$ (M)	OH^- (M)
Initial	0.490	0.450	0
Change	−x	+x	+x
Equilibrium	0.490 − x	0.450 + x	x

$$K_b = \frac{(0.450-x)(x)}{(0.490-x)} = 1.7 \times 10^{-9}$$

4. $K_b = \frac{(0.450)(x)}{(0.490)} = 1.7 \times 10^{-9}$

$x = 1.85 \times 10^{-9}$ M= $[OH^-]$

pOH = -log[OH-]

pOH = $-\log(1.85 \times 10^{-9}) = 8.73$

pH = 14.00 − pOH

pH = 14.00 - 8.73 = **5.27**

Q. Without the salt, the pH of pyridine (*aq*) would be:

a. < 5.27 **b.** > 5.27 **c.** = 5.27

The Common-Ion Effect

Q. Which substance, when added to 0.10 M HF, decreases the percent ionization of HF?

(a) NaCl
(b) NaF
(c) H_2O
(d) KNO_3

Answer: (b) NaF. F^- is the common ion and shifts $HF \rightleftharpoons H^+ + F^-$ to the left.

Q. Adding NH_4Cl to aqueous NH_3 will …

(a) increase the pH
(b) decrease the pH
(c) not change the pH
(d) make the solution neutral

Answer: (b). NH_4^+ shifts $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$ to the left, so $[OH^-]$ falls.

The Common-Ion Effect

Summary points

- ✓ A common ion suppresses the ionization of a weak acid or weak base (Le Châtelier).
- ✓ Weak acid + its salt (CH_3COOH / CH_3COO^-): $[H^+]$ decreases → pH increases.
- ✓ Weak base + its salt (NH_3 / NH_4^+): $[OH^-]$ decreases → pH decreases.
- ✓ In calculations, put the common ion's concentration in the "**Initial**" row of the ICE table.

Q. Why is the pH of a solution that is 0.10 M in HCl and 0.10 M in CH_3COOH essentially the same as that of 0.10 M HCl alone?

Answer: H^+ from HCl is a common ion that suppresses the ionization of CH_3COOH. The extra H^+ from CH_3COOH is negligible, so pH ≈ 1.00.

Buffers

Q. Your blood pH stays between **7.35** and **7.45**, even though your metabolism constantly produces acids. What keeps it so stable?

Answer: A buffer. The H_2CO_3 / HCO_3^- conjugate pair neutralises added acid or base, so the pH changes very little.

- Death may result if the blood pH falls below 6.8 or rises above 7.8!
  - When the pH falls below 7.35, the condition is called **acidosis**.
  - When the pH rises above 7.45, the condition is called **alkalosis**.
- Acidosis is the more common tendency because metabolism generates several acids in the body.

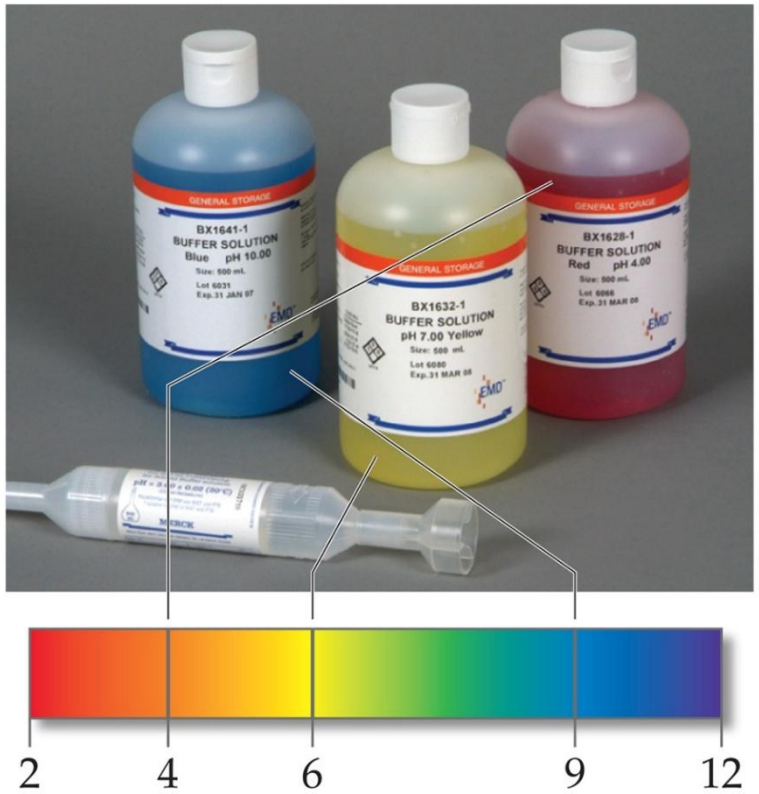
Q. Which organs regulate the pH of the carbonic acid-bicarbonate buffer system?

Buffers

Q. What are buffers (buffered solutions)?

- Solutions of a **weak conjugate acid–base pair**.
- Resist drastic pH changes when small amounts of strong acid or strong base are added to them.
- Contain relatively high concentrations (10^{-3} *M* or more) of both the acid and base at approximately equal concentrations.

Q. Will there be a neutralization reaction in buffers?



Standard buffers.
For laboratory work, prepackaged buffers at specific pH values can be purchased.

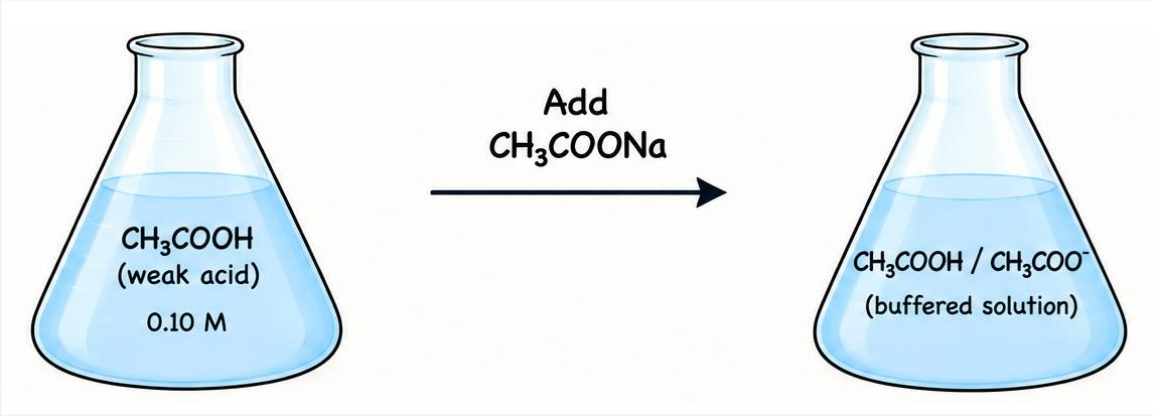
Buffers

Q. How to prepare buffers?

There are two ways to make a buffer.

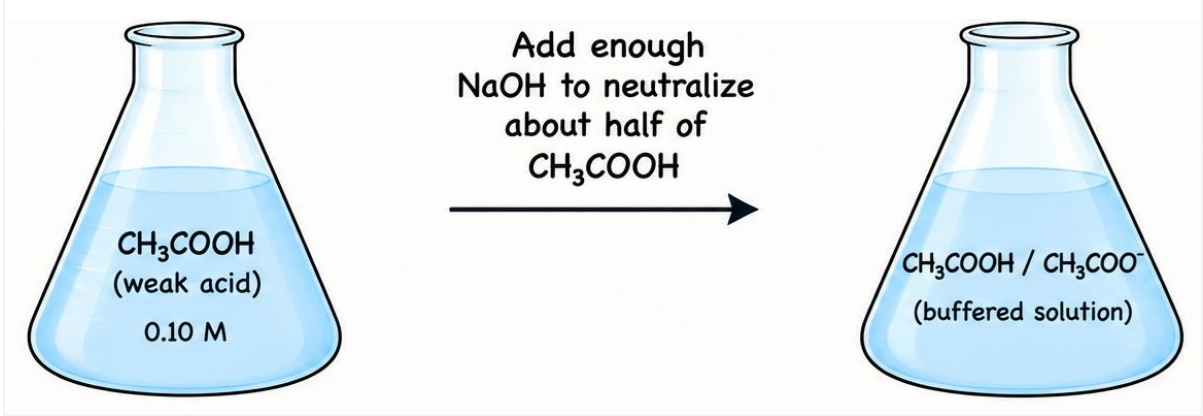
1. Direct mixing of a weak acid or a weak base with a **salt** of that acid or base.

Example: Adding CH_3COONa to a solution of CH_3COOH



2. Making the conjugate acid or base from a solution of a weak base or acid by the addition of a **strong** acid or base.

Example: Adding enough NaOH to neutralize about half of the CH_3COOH



$$CH_3COOH\,(aq) + OH^-\,(aq) \rightarrow CH_3COO^-\,(aq) + H_2O\,(l)$$

Buffers

Composition and Action of Buffers

Q. How do buffers resist changes in pH?

Consider: $HA\ (aq) \rightleftharpoons H^+\ (aq) + A^-\ (aq)$

Weak acid **Conjugate base**

$$K_a = \frac{[H^+]\,[A^-]}{[HA]} \qquad [H^+] = K_a \frac{[HA]}{[A^-]}$$

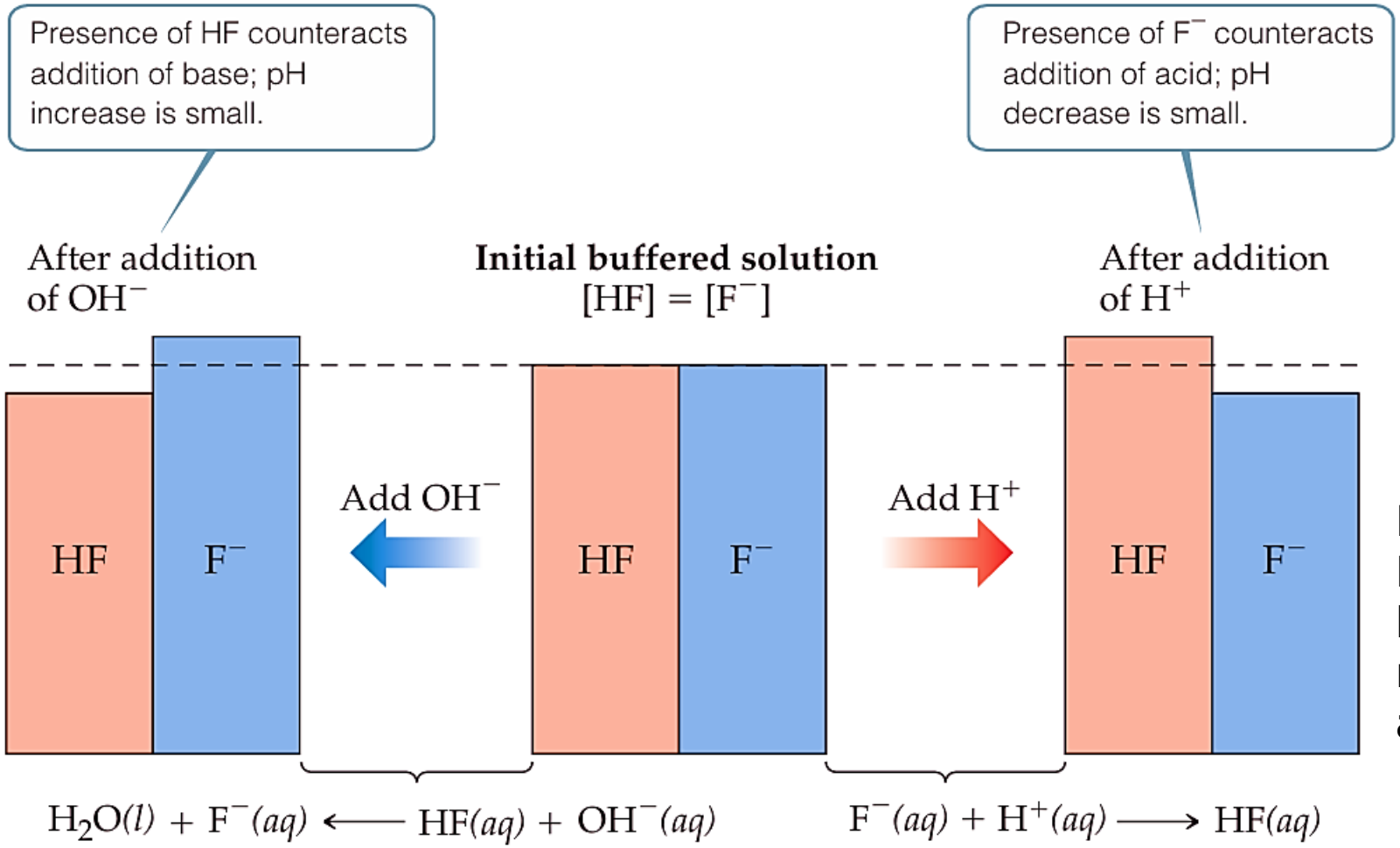
As long as the change in the ratio $[HA]/[A^-]$ is small, the change in pH will be small.

✓ The acid neutralizes added OH^- ions: $OH^-\ (aq) + HA\ (aq) \rightleftharpoons H_2O\ (l) + A^-\ (aq)$
added base

✓ The base neutralizes added H^+ ions: $H^+\ (aq) + A^-\ (aq) \rightleftharpoons HA\ (aq)$
added acid

Buffers

Composition and Action of Buffers



Buffer action. The pH of an HF/F- buffered solution changes by only a small amount in response to the addition of an acid or base.

Buffers

Calculating the pH of a buffer solution

- Because a buffer contains **a common ion**, its pH can be calculated using the same approach as the **common-ion effect**.

- Alternatively, the **Henderson–Hasselbalch equation** can be used.

$$\mathbf{pH} = \boldsymbol{pK_a} + \boldsymbol{log}\,\frac{[\boldsymbol{Base}]}{[\boldsymbol{Acid}]}$$

✓ [acid] and [base] refer to the equilibrium concentrations of the conjugate acid-base pair.

✓ When [base] = [acid], pH = pKa

Q. Derive the Henderson-Hasselbalch equation from the following.

$HA\ (aq) \rightleftharpoons H^+\ (aq) + A^-\ (aq)$

$$[\boldsymbol{H^+}] = \boldsymbol{K_a}\,\frac{[\boldsymbol{HA}]}{[\boldsymbol{A^-}]}$$

Buffers

Q. What is the pH of a buffer that is 0.12 M in lactic acid ($CH_3CH(OH)COOH$, $K_a = 1.4 \times 10^{-4}$) and 0.10 M in sodium lactate ($CH_3CH(OH)COONa$)?

$$\boldsymbol{pH = pK_a + log\frac{[Base]}{[Acid]}}$$

$$\boldsymbol{pH = -logK_a + log\frac{(0.10)}{(0.12)}}$$

$$\boldsymbol{pH = 3.85 + (-0.08) = 3.77}$$

$pKa = -\log(1.4 \times 10^{-4})$

$pKa = -[\log(1.4) + \log(10^{-4})]$

$= -[(0.1461) + (-4)]$

$= 3.85$

Q. For the above question, apply the ICE table method from the common ion effect and determine the pH value.

Buffers

Q. A buffer made with sodium acetate (CH_3COONa) and acetic acid (CH_3COOH, K_a = 1.80 x 10^{-5}) has a pH of 3.98. What is the ratio of the equilibrium concentration of sodium acetate to that of acetic acid? (**Ans**. 1 : 0.172)

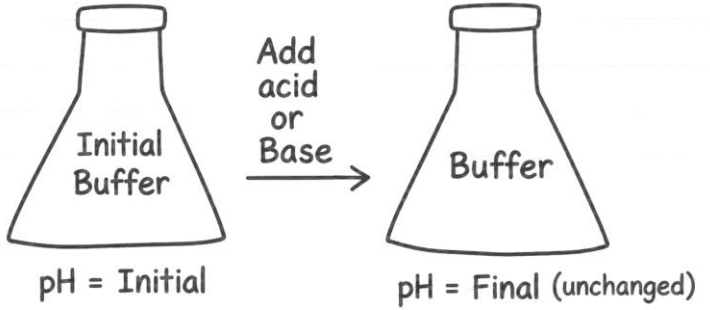
$$pH = pK_a + log\frac{[CH_3COONa]}{[CH_3COOH]};\quad \frac{[CH_3COONa]}{[CH_3COOH]} = 10^{pH-pK_a} = 10^{3.98-[-\log(1.80\times10^{-5})]} = 10^{3.98-4.745} = 10^{-0.765} = 0.172$$

Q. How many moles of NH_4Cl must be added to 2.0 L of 0.10 M NH_3 (K_b = 1.8 x 10^{-5}) to form a buffer whose pH is 9.00? (Assume that the addition of NH_4Cl does not change the volume of the solution.) (***Ans***. 0.36 mol NH_4Cl).

Q. Calculate the number of grams of ammonium chloride that must be added to 2.00 L of a 0.500 M ammonia solution to obtain a buffer of pH = 9.20. Assume the volume of the solution does not change as the solid is added. K_b for ammonia is 1.80 x 10^{-5}.

Buffers

Addition of Strong Acids or Bases to Buffers



Q. How does the pH of the buffer respond to the addition of a strong acid or a strong base?

- Strong acids and bases react essentially to completion with weak bases and weak acids, respectively (**Neutralization reactions**).

- Added strong acid or base is consumed by the buffer as long as its buffering capacity is not exceeded.

$$\mathbf{HA\ (aq) \rightleftharpoons H^+\ (aq) + A^-\ (aq)}$$

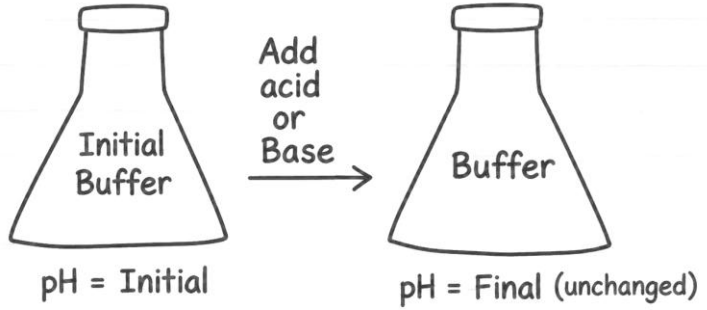
✓ Adding strong acid ($H^+ + A^- \rightarrow HA$): [HA] increases, while [A^-] decreases.

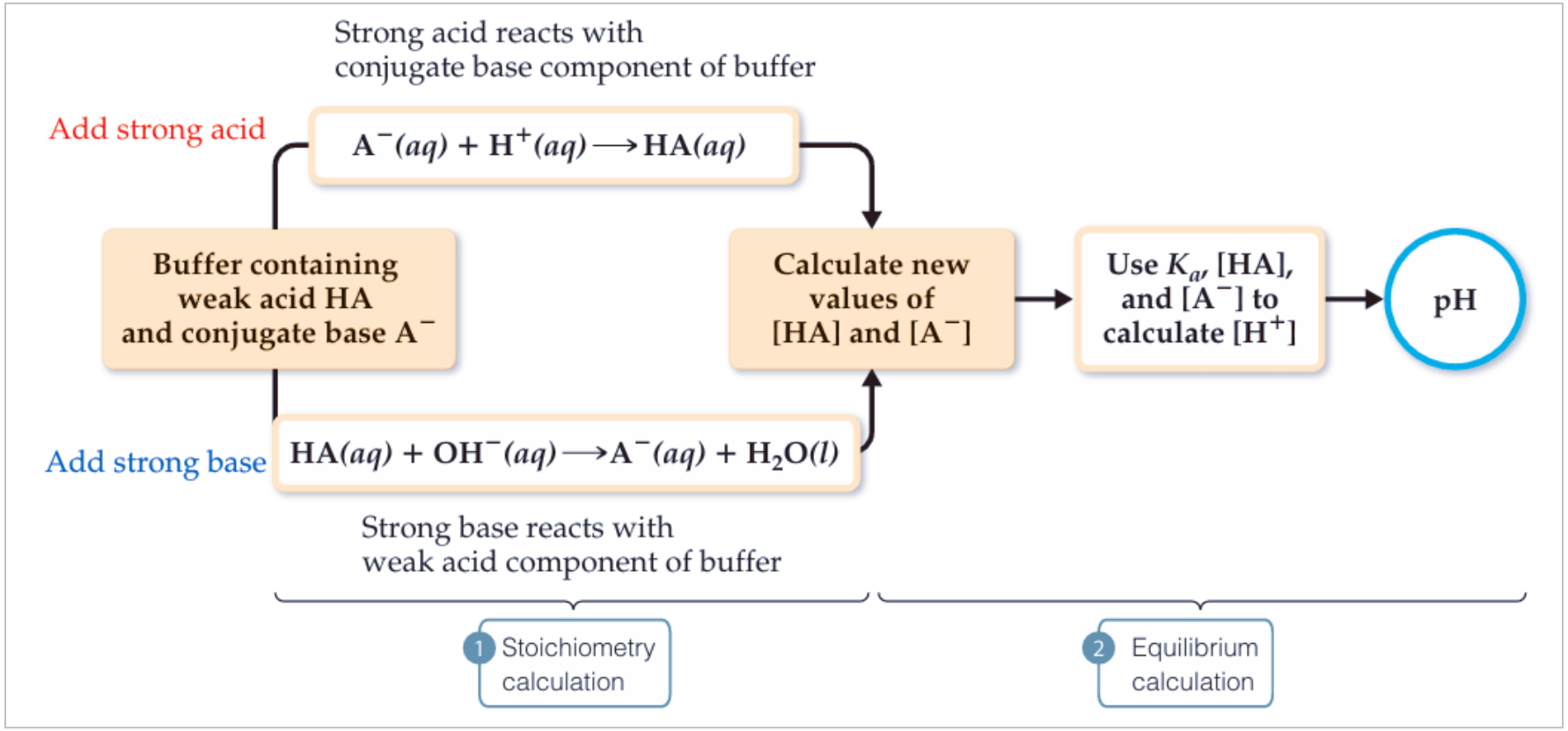
✓ Adding strong base ($OH^- + HA \rightarrow A^- + H_2O$). [HA] decreases, [$A^-$] increases.

Buffers

Addition of Strong Acids or Bases to Buffers

Q. How does the pH of the buffer respond to the addition of a strong acid or a strong base?





$$pH = pK_a + log\frac{[Base]}{[Acid]}$$

Henderson–Hasselbalch equation

Calculating the pH of a buffer after the addition of a strong acid or strong base.

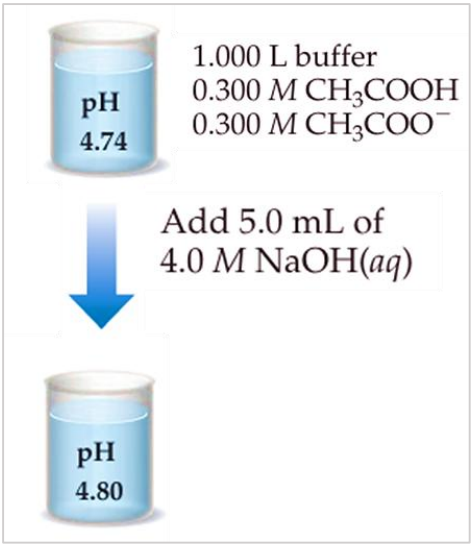
Buffers

Q. A buffer made by adding 0.300 mol CH_3COOH (K_a = 1.80 x 10^{-5}) and 0.300 mol CH_3COONa to enough water to make 1.00 L of solution has a pH of 4.74. Calculate the pH after 5.0 mL of 4.0 M NaOH(aq) solution is added.

$$CH_3COOH(aq) + OH^-(aq) \rightarrow H_2O(l) + CH_3COO^-(aq)$$

Step 1. *Stoichiometry Calculation (**find remaining moles after the reaction**):*

	CH_3COOH	OH^-	H_2O	CH_3COO^-
Before reaction, **mol**	0.300	0.020	—	0.300
Change (limiting reactant), **mol**	-0.020	**-0.020**	—	+0.020
After reaction, **mol**	0.280	0	—	0.320



Step 2. Equilibrium Calculation: $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$

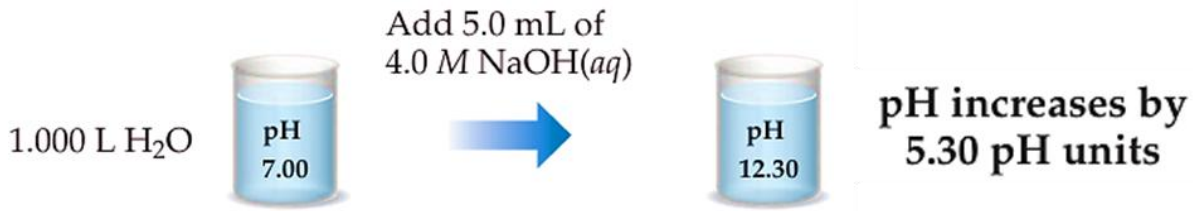
$$pH = pK_a + log\frac{[Base]}{[Acid]}$$

$$pH = 4.74 + log\frac{0.320\ mol/1.005\ L}{0.280\ mol/1.005\ L} = 4.80$$

$V_{final} = V_{Initial} + V_{NaOH}$
$= 1.005$ L

Buffers

Q. Determine the pH of a solution made by adding 0.020 mol of NaOH to 1.000 L of pure water. Compare the pH change with the value on the previous slide.



Q. A buffer contains 0.30 mol of propanoic acid (C_2H_5COOH, $K_a = 1.3 \times 10^{-5}$) and 0.25 mol of potassium propanoate (C_2H_5COOK) in 1.80 L.

(a) What is the pH of this buffer?
(b) What is the pH of the buffer after the addition of 0.10 mol of NaOH?
(c) What is the pH of the buffer after the addition of 0.10 mol of HCl?

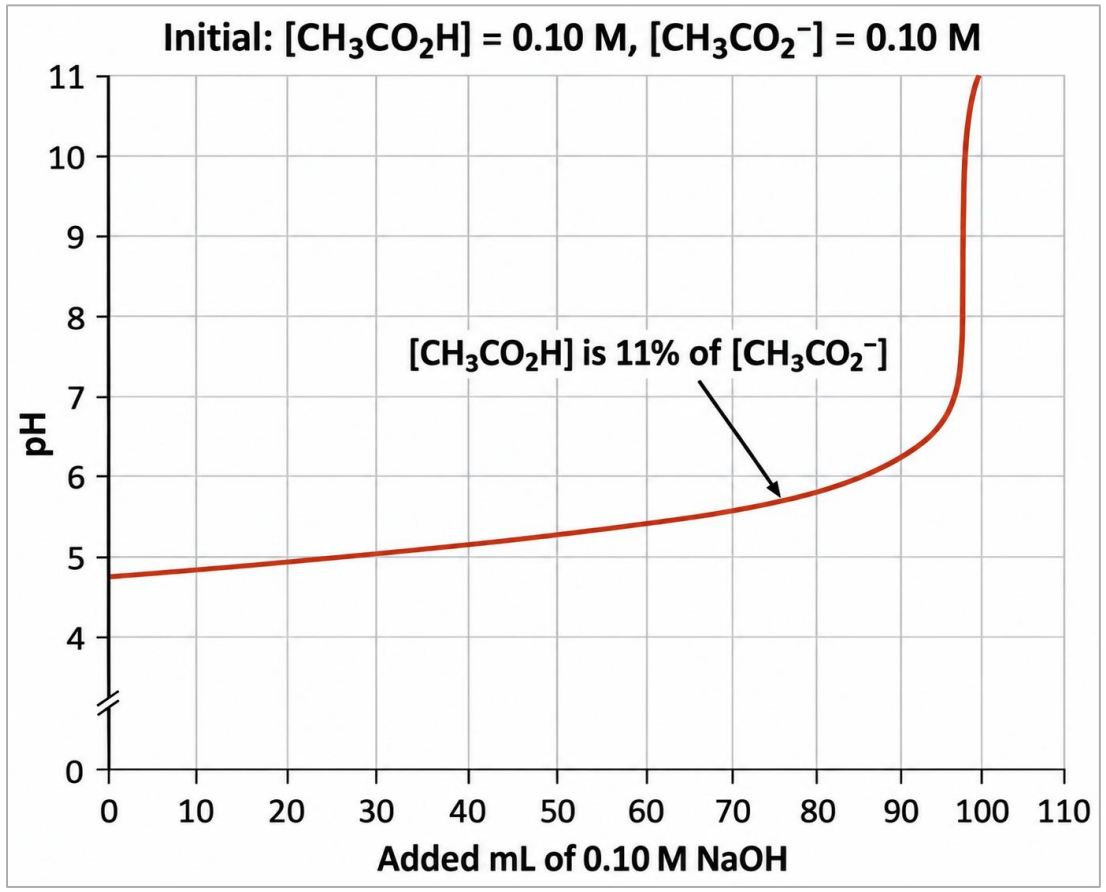
Q. Which of the following solutions is a buffer?

(a) A solution made by mixing 50 mL of 0.200 M HCOOH and 250 mL of 0.200 M KOH.
(b) A solution made by mixing 50 mL of 0.200 M HCOOH and 25 mL of 0.200 M KOH.
(c) A solution made by mixing 50 mL of 0.200 M HCOOH and 25 mL of 0.200 M HNO_3.
(d) A solution made by mixing 50 mL of 0.200 M potassium formate (HCOOK) and 25 mL of 0.200 M KNO_3.

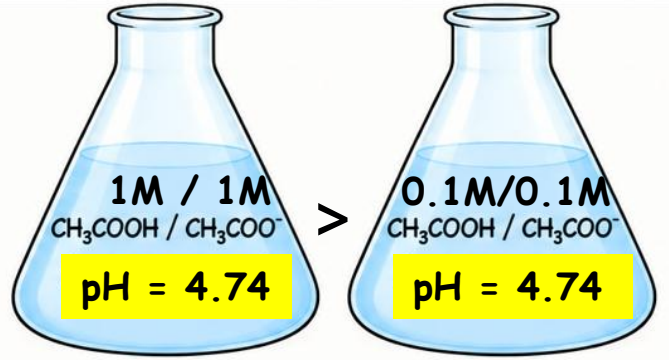
Buffers

Buffer Capacity and pH range

Q. How far can a buffer resist changes in pH?



- **Buffer capacity:** the amount of strong acid or base a buffer can neutralize before its pH changes appreciably.
- Buffer capacity depends on the amounts of weak acid and conjugate base present.
- More concentrated buffers have greater capacity.

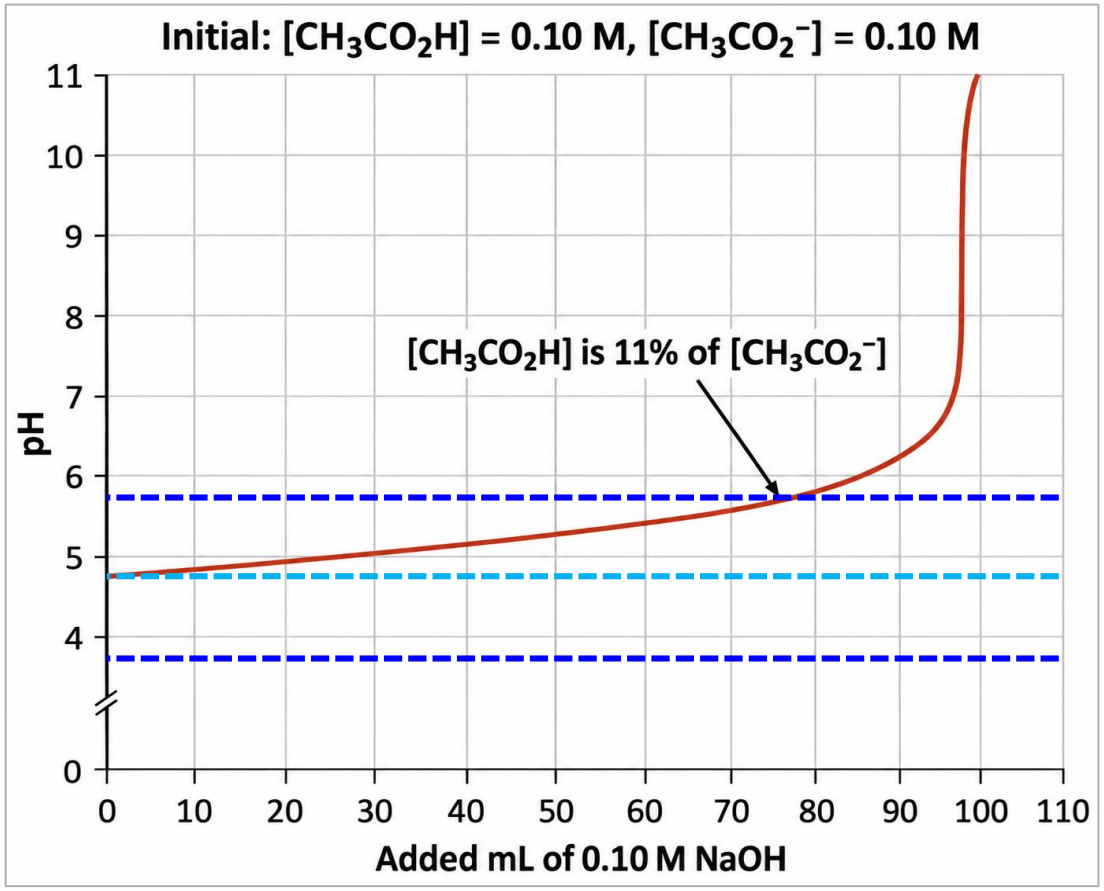


Relative buffer capacity

Buffers

Buffer Capacity and pH range

Q. How far can a buffer resist changes in pH?



- **The pH range of a buffer**: the pH range over which the buffer acts effectively.
- **Optimal pH**: A buffer works best when the weak acid and conjugate base concentrations are about the same (i.e. pH = pKa)

$$pH = pK_a + log\frac{[Base]}{[Acid]}$$

- **Effective pH**: A buffer system is most effective at stabilizing pH within ±1 pH unit of its pKa.

Buffers

Q. Which mixture forms a buffer solution?

(a) $HCl + NaCl$
(b) $HNO_2 + NaNO_2$
(c) $NaOH + NaCl$
(d) $HNO_3 + KNO_3$

Answer: (b). A weak acid (HNO_2) together with its conjugate base (NO_2^-).

Q. You need a buffer with pH = 4.8. Which weak acid (with its salt) is the best choice?

(a) $HCOOH$, $pK_a = 3.74$
(b) CH_3COOH, $pK_a = 4.74$
(c) $H_2PO_4^-$, $pK_a = 7.21$
(d) NH_4^+, $pK_a = 9.25$

Answer: (b). Choose pK_a closest to the target pH (effective range $\approx pK_a \pm 1$).

Buffers

Summary points

- ✓ Buffer = weak acid + conjugate base (or weak base + conjugate acid) in appreciable amounts.
- ✓ HA neutralises added OH^-; A^- neutralises added H^+.
- ✓ $pH = pK_a + \log([base]/[acid])$; when [base] = [acid], $pH = pK_a$.
- ✓ Strong acid/base added: stoichiometry first (moles), then Henderson–Hasselbalch.
- ✓ Capacity increases with concentration; effective pH range ≈ $pK_a \pm 1$.

Q. A buffer is 0.20 M NH_3 and 0.20 M NH_4Cl (pK_a of NH_4^+ = 9.25). What is its pH? What changes if both concentrations are doubled?

Answer: pH = 9.25 in both cases. Doubling the concentrations increases the buffer capacity, not the pH.

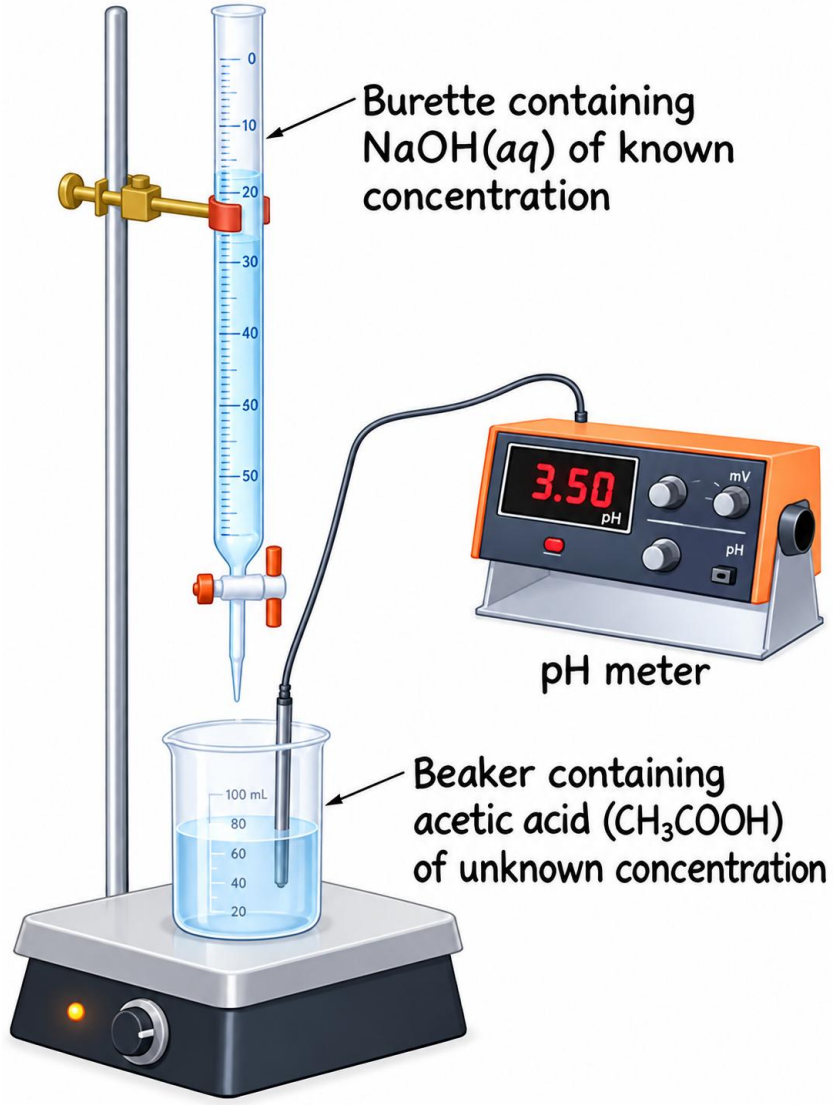
$$\boldsymbol{pH = pK_a + log\frac{[Base]}{[Acid]}}$$

Acid-Base Titrations

Q. If you slowly add NaOH to an acetic acid solution, how will the pH of the solution change? Consider drawing a graph of added NaOH volume (mL) vs. pH Change.

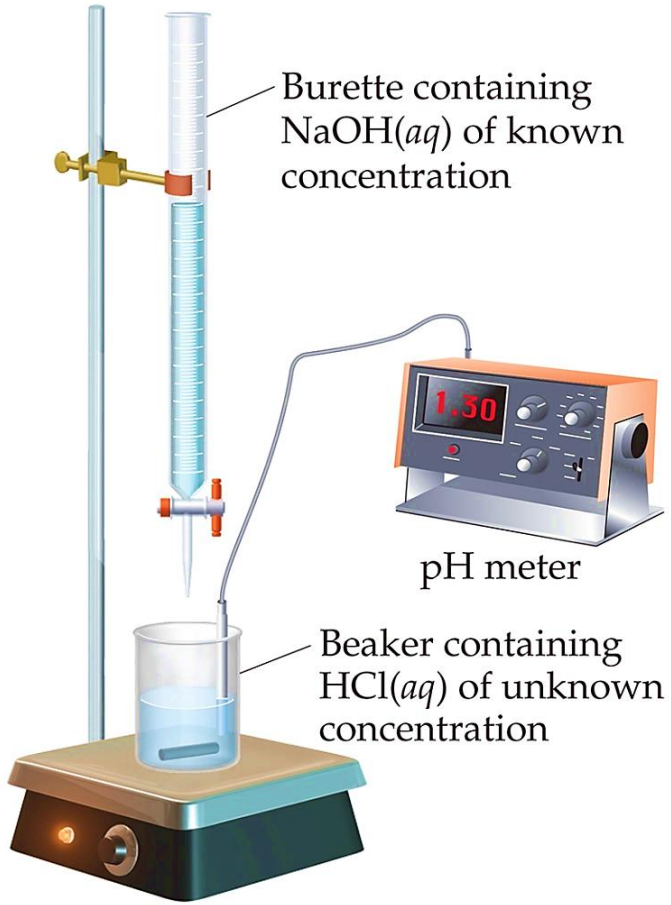
- ✓ When will the solution become acidic?
- ✓ When will the solution become neutral?
- ✓ When will the solution become basic?

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$



Acid-Base Titrations

Q. What are titrations? And their applications?

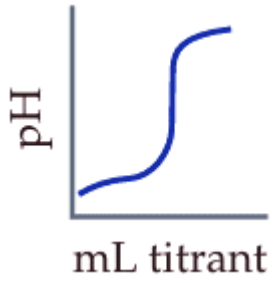


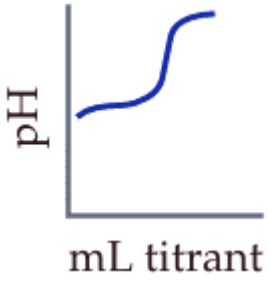
Measuring pH during a titration

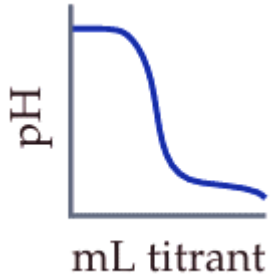
- **Titration**: A procedure in which one reactant is gradually added to another while monitoring the reaction.
- **Main purposes**: Determine the concentration of a reactant or the equilibrium constant of a reaction.
- **Acid–base titration**: A solution of known concentration is added gradually to an acid or base of unknown concentration.
- **Equivalence point**: The point at which stoichiometrically equivalent amounts of acid and base have reacted.

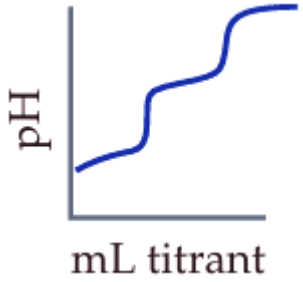
Acid-Base Titrations

Q. What are titrations? And their applications?





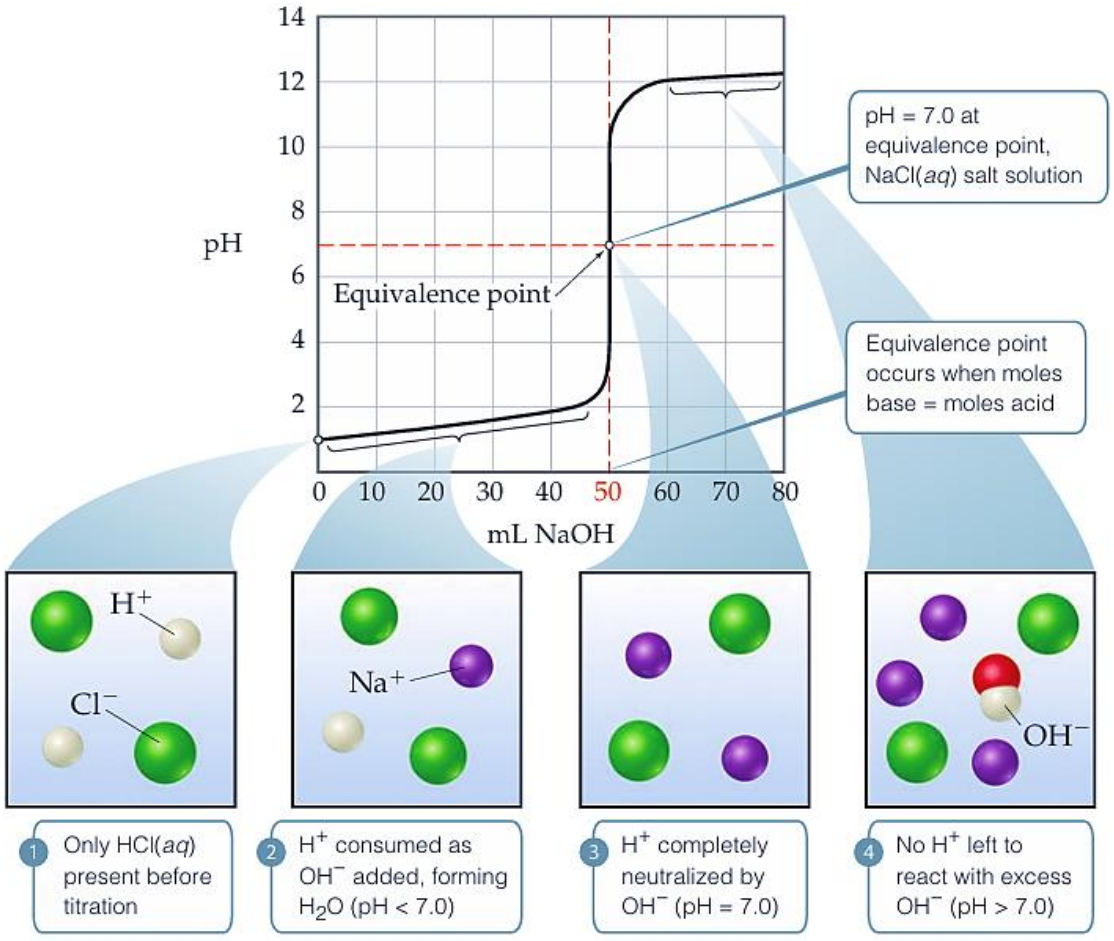




- A graph of pH versus the volume of **titrant added** is called **a titration curve**.
- The shape of a titration curve helps identify the equivalence point and select a suitable indicator.
- Titration curves can also be used to determine K_a of a weak acid or K_b of a weak base.
- Common types include:
  - ✓ Strong acid–strong base
  - ✓ Weak acid–strong base
  - ✓ Polyprotic acid–strong base titrations

Acid-Base Titrations

Strong Acid–Strong Base Titrations



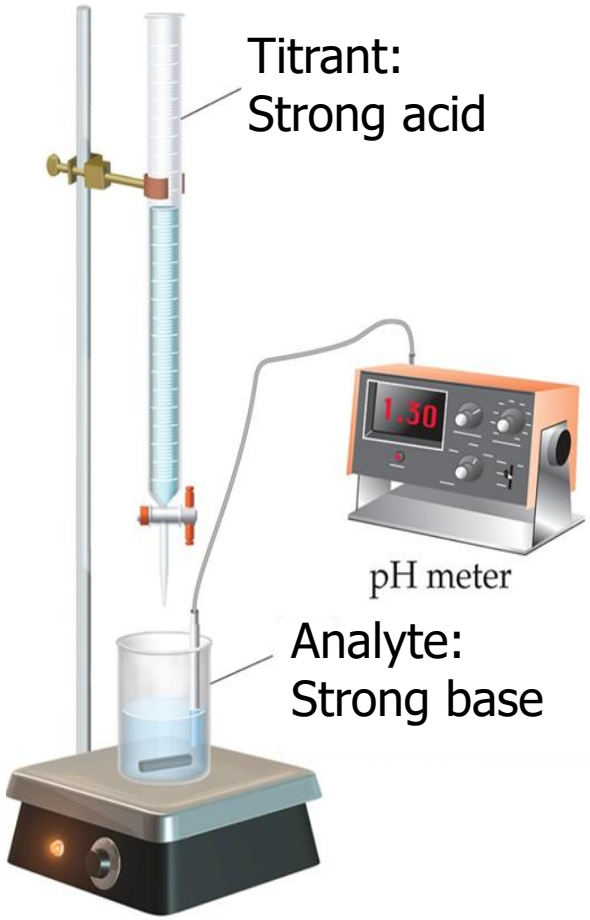
Titration of a strong acid with a strong base.

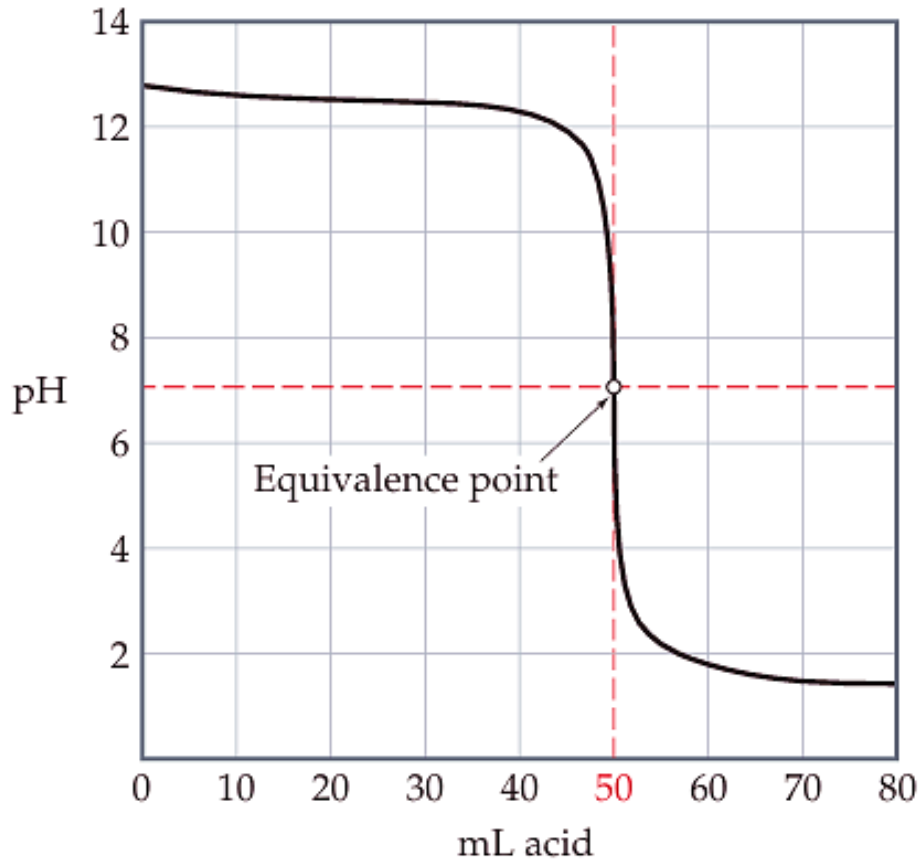
1. **Initial pH**: Determined by the initial strong acid concentration (pH < 7).
2. **Before the equivalence point**: pH increases as NaOH neutralizes HCl. It rises slowly at first, then rapidly near the equivalence point. The pH depends on the remaining **unreacted acid**.
3. **Equivalence point**: Equal moles of HCl and NaOH have reacted, leaving NaCl in water. pH = 7.00.
4. **After the equivalence point:** pH is determined by the concentration of **excess NaOH** remaining in the solution.

The pH curve for titration of 50.0 mL of a 0.100 M solution of hydrochloric acid with a 0.100 M solution of NaOH(aq). For clarity, water molecules have been omitted from the molecular art.

Acid-Base Titrations

Strong Base–Strong Acid Titrations





▲ **Figure 17.8 Titration of a strong base with a strong acid.** The pH curve for titration of 50.0 mL of a 0.100 *M* solution of a strong base with a 0.100 *M* solution of a strong acid.

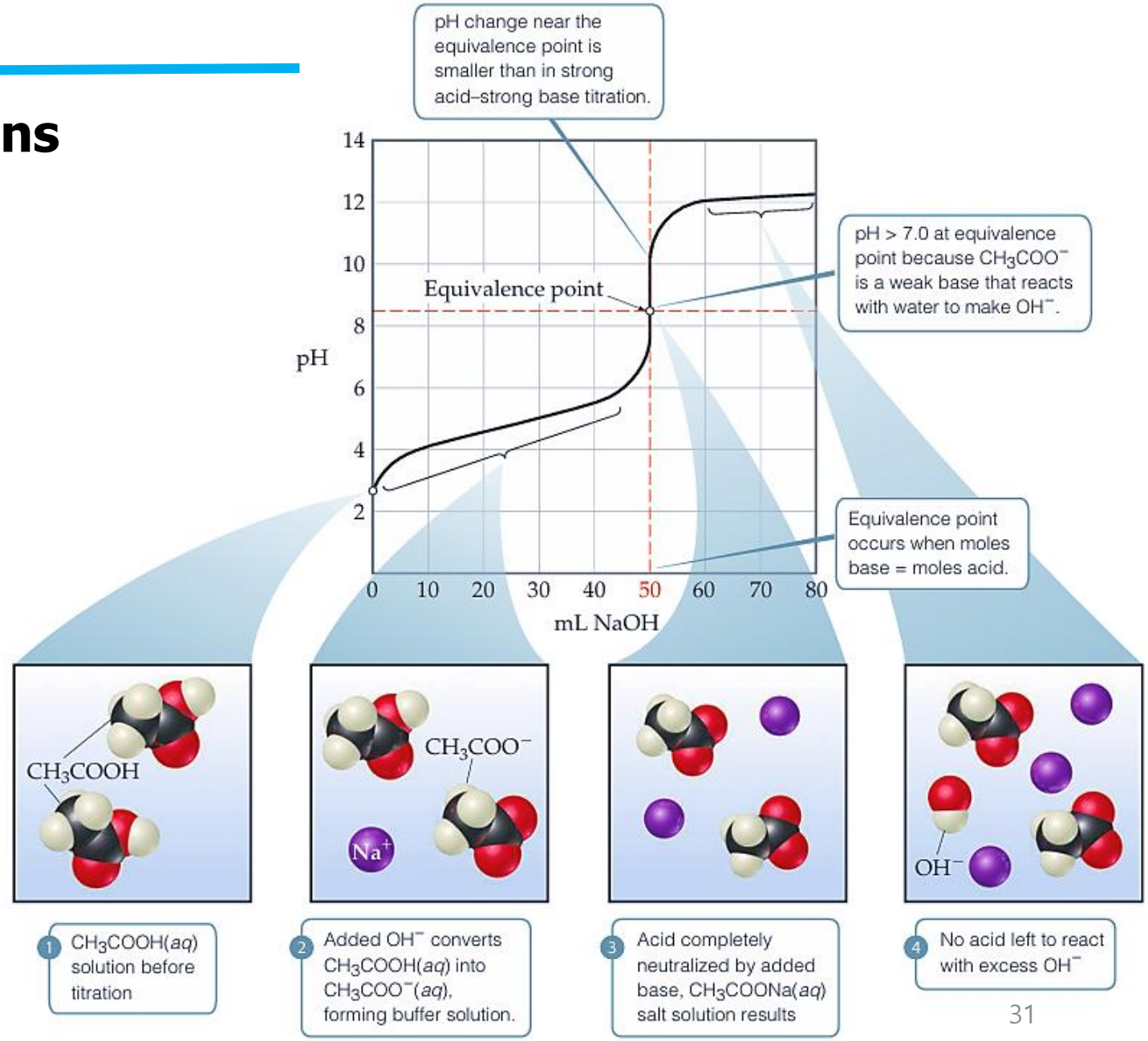
Q. Explain the following regions for **strong base–strong acid titrations** and how the pH is determined.

1. Initial pH (pH > 7)
2. Before the equivalence point (pH > 7)
3. At the equivalence point (pH = 7)
4. After the equivalence point (pH < 7)

Acid-Base Titrations

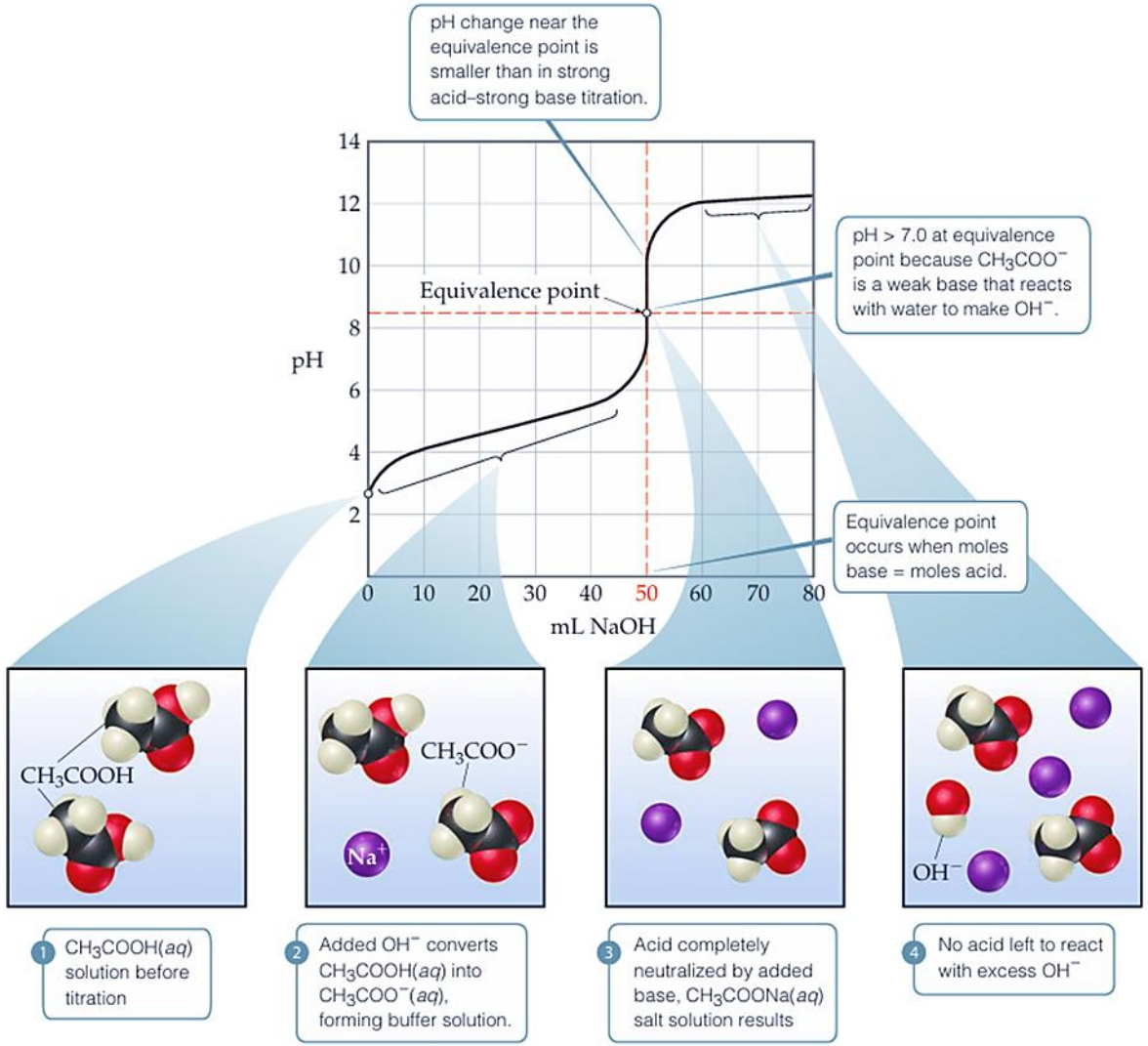
Weak Acid–Strong Base Titrations

Titration of a weak acid with a strong base. The pH curve for the titration of 50.0 mL of a 0.100 M solution of acetic acid with a 0.100 M solution of NaOH(aq). For clarity, water molecules have been omitted from the molecular art.



Acid-Base Titrations

Weak Acid–Strong Base Titrations



1. **Initial pH**: Calculate using K_a and **ICE table**. For 0.100 M CH_3COOH, pH = 2.89.

2. **Before the equivalence point**: Added NaOH partially neutralizes CH_3COOH, forming its conjugate base, CH_3COO^- (The solution forms a buffer).

$CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$

$$pH = pK_a + log\frac{[Base]}{[Acid]}$$

pH calculation involves two steps (See next slide)

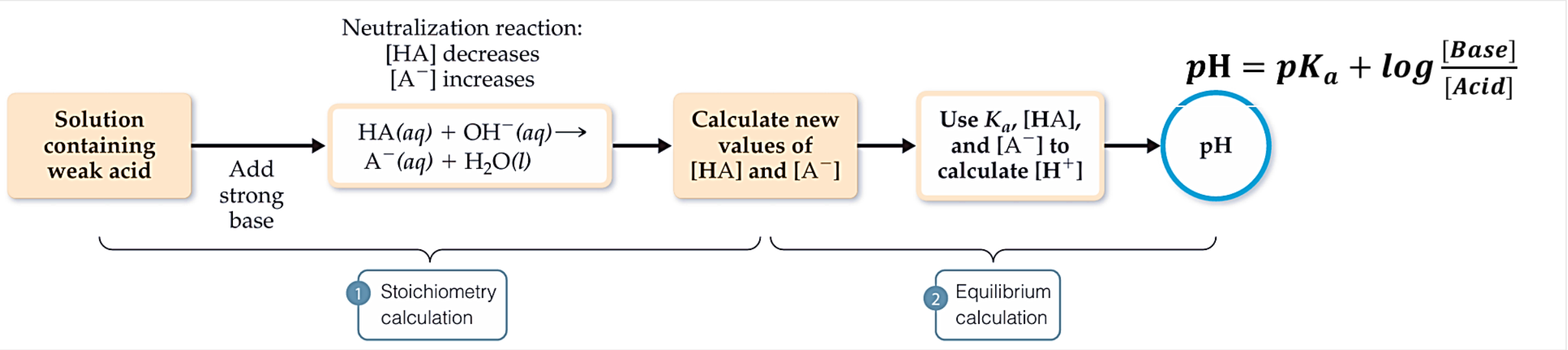
Acid-Base Titrations

Weak Acid–Strong Base Titrations

2. Before the equivalence point…

pH calculation involves two steps

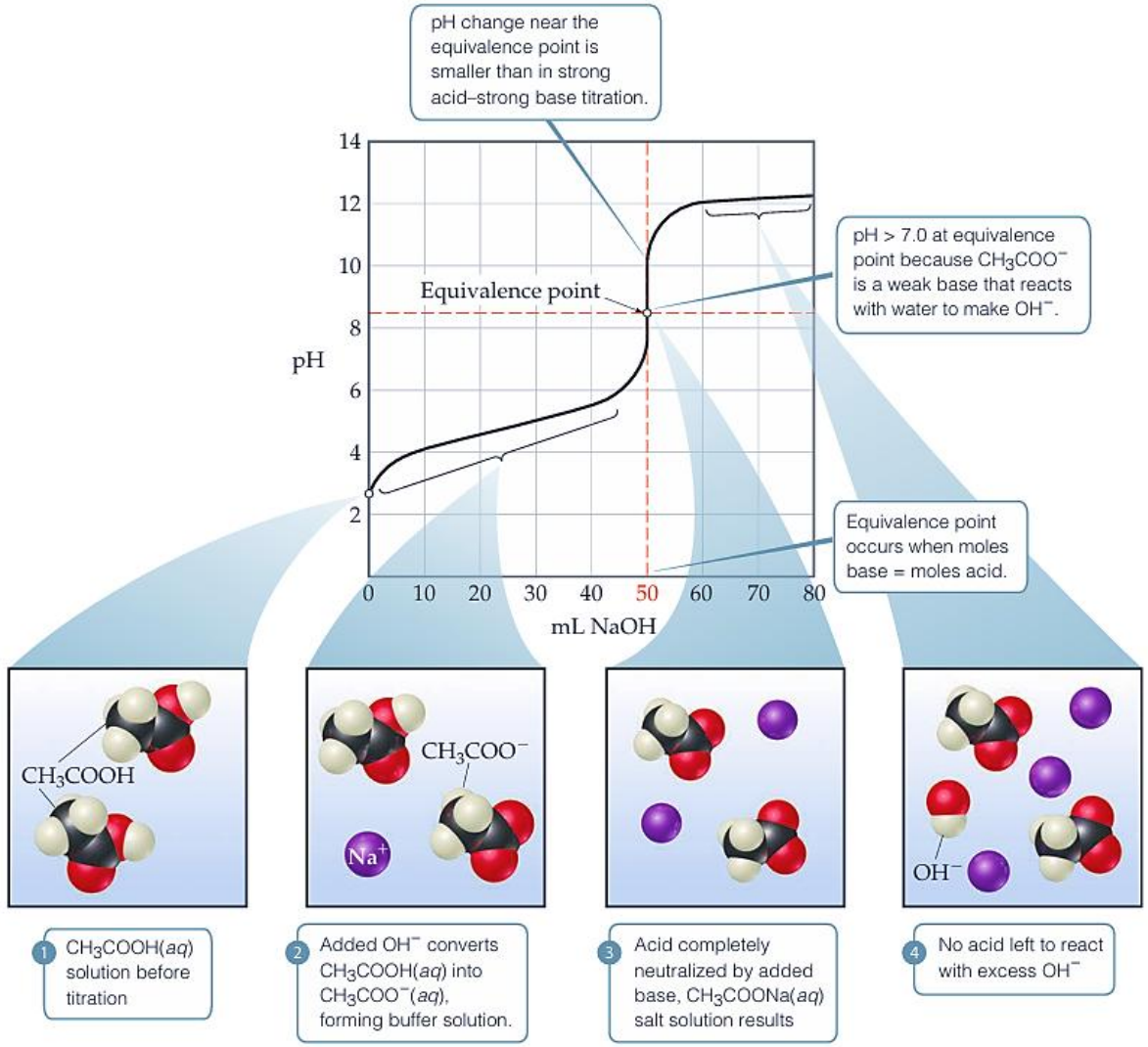
- ✓ **Step 1:** Calculate the remaining CH_3COOH and the formed CH_3COO^- using stoichiometry.
- ✓ **Step 2:** Calculate pH using the Henderson–Hasselbalch equation.



Procedure for calculating pH when a weak acid is partially neutralized by a strong base.

Acid-Base Titrations

Weak Acid–Strong Base Titrations



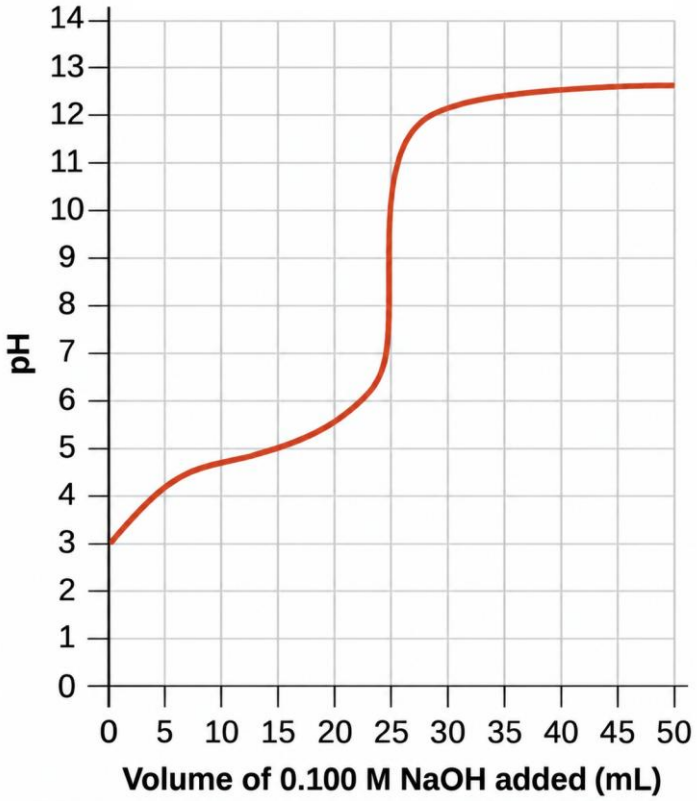
3. Equivalence point: Equal moles of CH_3COOH and NaOH have reacted, forming CH_3COONa. The CH_3COO^- ion reacts with water to produce OH^-, so **pH > 7.**

4. After the equivalence point: Excess NaOH determines the pH; OH^- from acetate hydrolysis is negligible. At 51.0 mL of 0.100 M NaOH added to 50.0 mL of 0.100 M CH_3COOH, pH = 11.00, the same as for a strong acid–strong base titration under these conditions.

Acid-Base Titrations

Q. The following titration curve is obtained from the titration of 25.00 mL of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) with 0.100 M NaOH.

$$CH_3COOH\ (aq) + OH^-\ (aq) \rightarrow CH_3COO^-\ (aq) + H_2O\ (l)$$



a. Determine the initial pH of the solution. (***Ans. 2.87***, *Use an ICE table and K_a to find $[H_3O^+]$ from a weak acid dissociation)*

b. **What is the pH after the addition of 12.50 mL of the NaOH solution (half the equivalence point)?**

c. **What is the pH after the addition of 25.00 mL of the NaOH solution (Equivalence point)?**

d. What is the pH after the addition of 37.50 mL of the NaOH solution (past the equivalence point)? (***Ans. 12.30***; *Find what is left in the beaker and determine pH from pOH.)*

Acid-Base Titrations

Q. titration of 25.00 mL of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) with 0.100 M NaOH.

$$CH_3COOH\ (aq) + OH^-\ (aq) \rightarrow CH_3COO^-\ (aq) + H_2O\ (l)$$

b. pH after 12.50 mL NaOH solution is added (half the equivalence point)?

1. **Stoichiometry (1:1)**

$$M = \frac{n}{V\ (L)}\ ;\ \mathbf{n = M \times V}$$

	CH_3COOH (aq)	OH^- (aq)	CH_3COO^- (aq)
S (Start), mol	0.002500	**0.001250**	0
C (Change), mol	−0.001250	−0.001250	+0.001250
F (Finish), mol	0.001250	0	0.001250

What's left in the beaker?
Equal moles of CH_3COOH and CH_3COO^-

✓ **Total volume = 25.00 mL + 12.5 mL = 37.50 mL = 0.0375 L**

2. $$pH = pK_a + log\frac{[Base]}{[Acid]}$$

$pK_a = -\log(1.8 \times 10^{-5}) = \mathbf{4.74}$

$$pH = 4.74 + log\frac{(0.001250)}{(0.001250)}$$

$pH = 4.74 + \log(1)$

$pH = 4.74 + 0 = \mathbf{4.74}$

Acid-Base Titrations

Q. titration of 25.00 mL of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) with 0.100 M NaOH.

$$CH_3COOH\ (aq) + OH^-\ (aq) \rightarrow CH_3COO^-\ (aq) + H_2O\ (l)$$

c. pH after the addition of 25.00 mL of the NaOH solution?

1. Stoichiometry (1:1) $M = \frac{n}{V\ (L)}$; n = M × V

	CH_3COOH (aq)	OH^- (aq)	CH_3COO^- (aq)
Start, mol	0.002500	0.002500	0
Change, mol	−0.002500	−0.002500	+0.002500
Finish, mol	0	0	**0.002500**

What's left in the beaker?
Excess CH_3COO^-

✓ Total volume = 25.00 mL + 25 mL = 50 mL = 0.05 L)

2. $[CH_3COO^-] = \frac{(0.002500\ mol)}{(0.0500\ L)} = 0.0500\ M$

$CH_3COO^-\ (aq) + H_2O\ (l) \rightarrow CH_3COOH\ (aq) + OH^-(aq)$

	CH_3COO^- (aq)	CH_3COOH (aq)	OH^- (aq)
I	0.0500	0	0
C	−x	+x	+x
E	0.0500 - x	x	x

3. $K_b = \frac{[OH^-]\,[CH3COOH]}{[CH_3COO^-]}$ $K_b = \frac{Kw}{Ka} = 5.6 \times 10^{-10}$

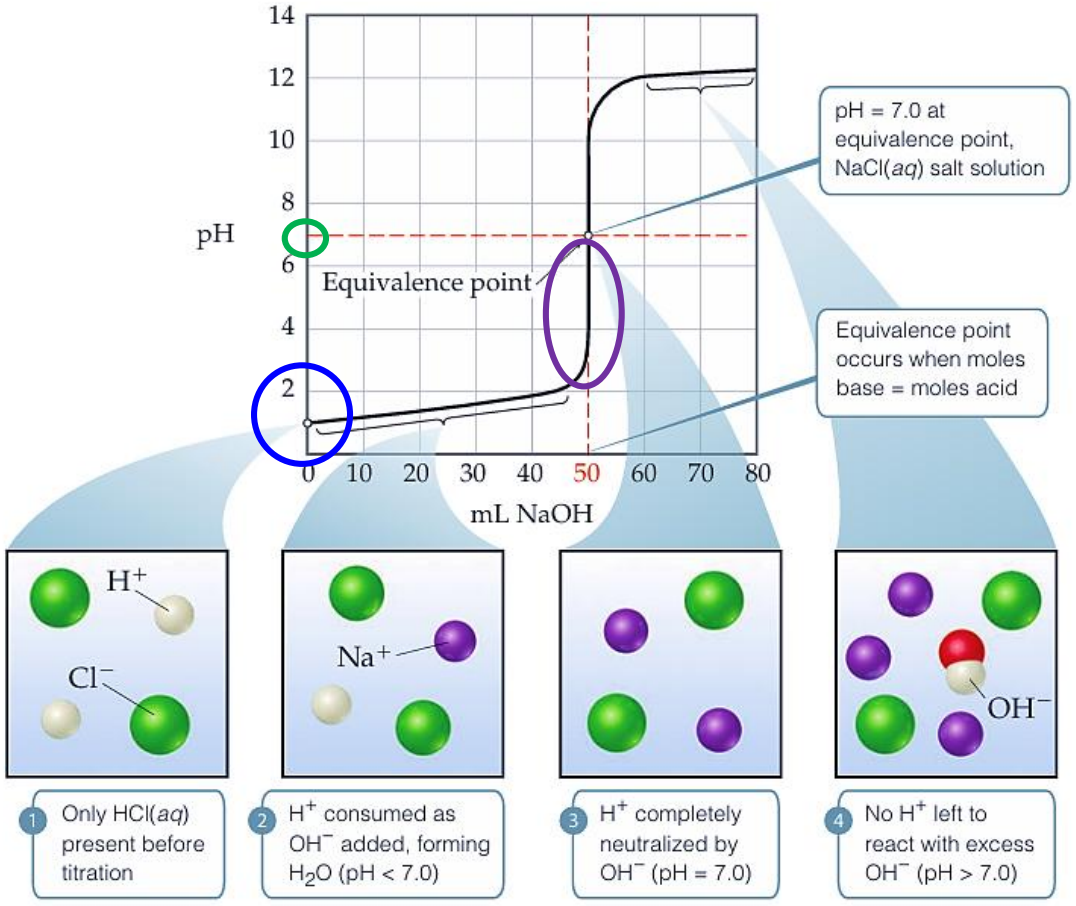
$5.6 \times 10^{-10} = \frac{x^2}{(0.0500-x)} = \frac{x^2}{(0.0500)}$; $x = 5.3 \times 10^{-6} = [OH^-]$

4. pOH = -log[OH^-] = -log(5.3×10^{-6}) = 5.28

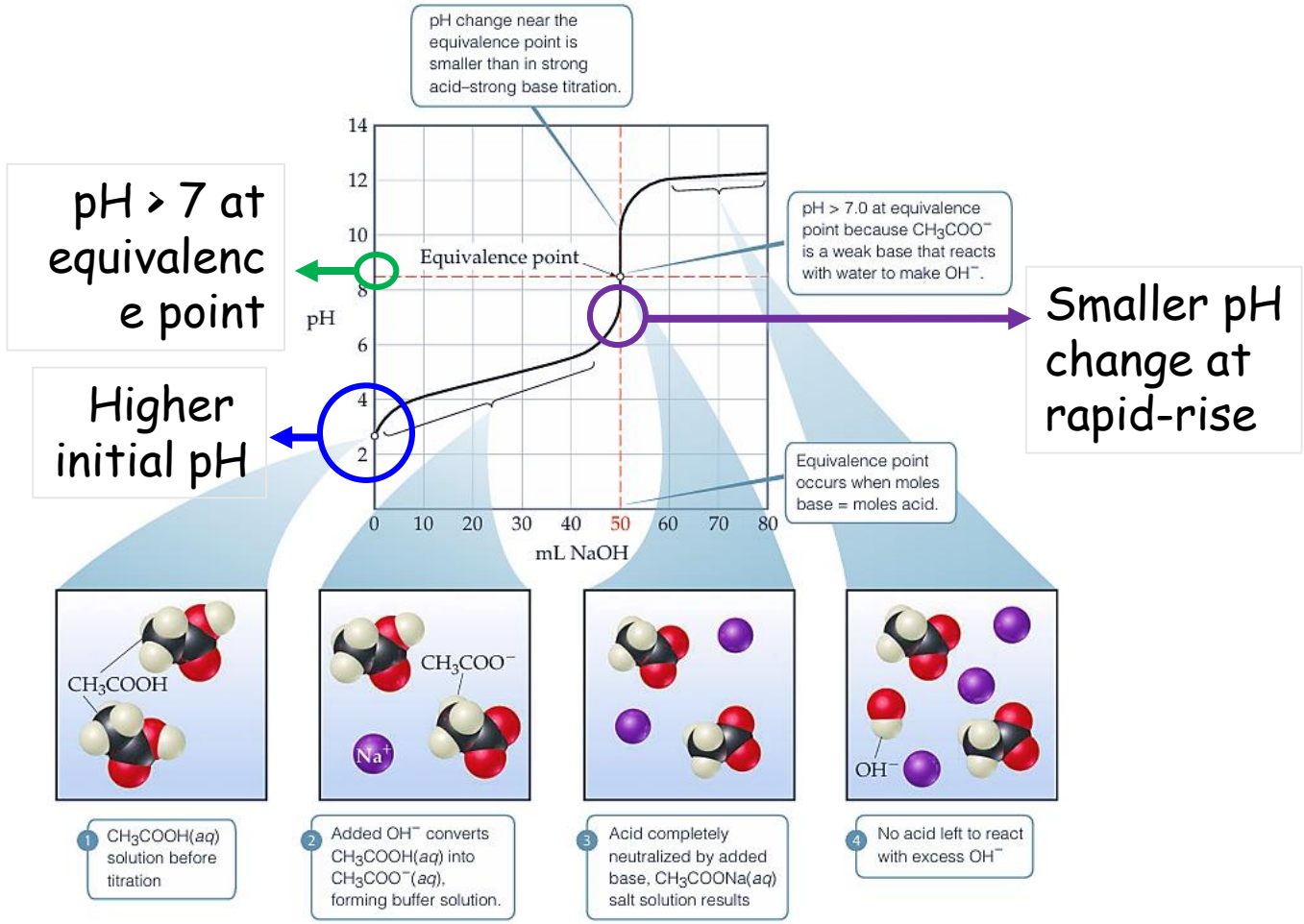
pH = 14.00 - pOH = 14.00 - 5.28 = **8.72**

Acid-Base Titrations

Q. How does a weak acid–strong base titration curve differ from that of a strong acid–strong base?



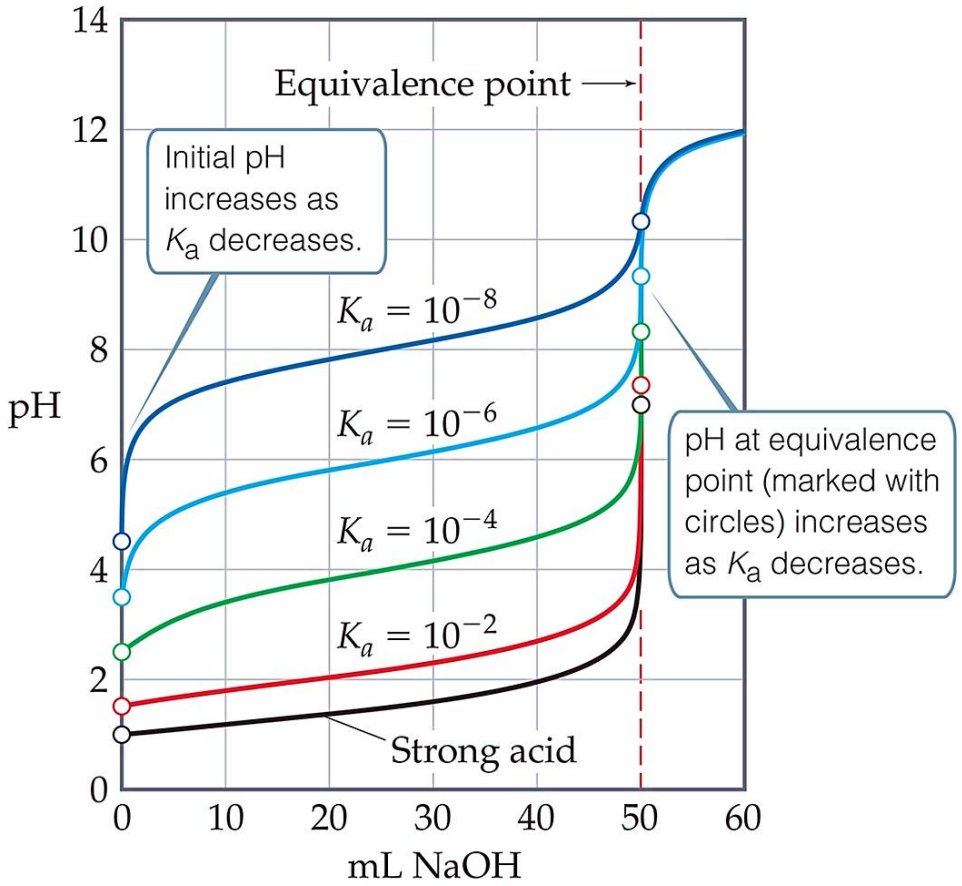
Strong acid-Strong base



Weak acid-Strong base

Acid-Base Titrations

Q. How does a weak acid–strong base titration curve differ from that of a strong acid–strong base?



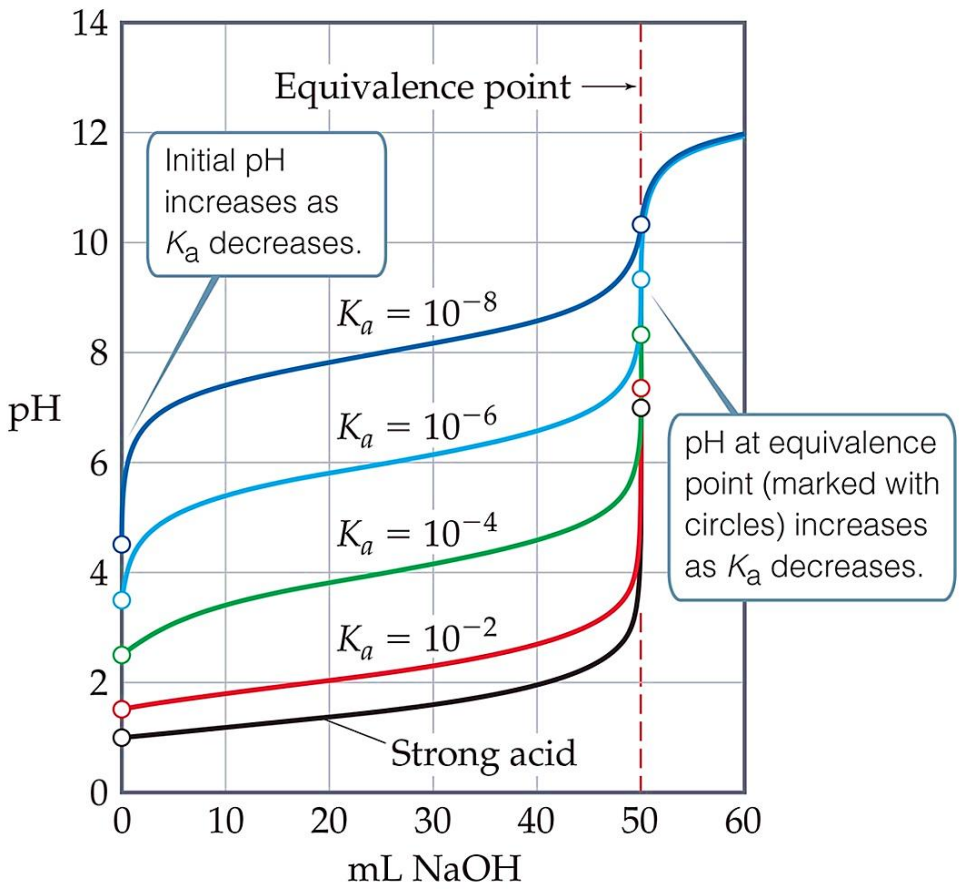
A set of curves showing the effect of acid strength on the characteristics of the titration curve when a weak acid is titrated by a strong base. Each curve represents titration of 50.0 mL of 0.10 M acid with 0.10 M NaOH.

As K_a becomes smaller:

- Initial pH and equivalence-point pH increase.
- The pH change near the equivalence point becomes less pronounced.
- For pKa ≥10, the equivalence point is difficult to identify due to the small gradual pH change.

Acid-Base Titrations

Q. How does a weak acid–strong base titration curve differ from that of a strong acid–strong base?



A set of curves showing the effect of acid strength on the characteristics of the titration curve when a weak acid is titrated by a strong base. Each curve represents titration of 50.0 mL of 0.10 M acid with 0.10 M NaOH.

Q. How to measure the pKa of a weak acid from pH titration experiments?

- Identify the volume of strong base required to reach the equivalence point.
- At half the equivalence-point volume, [HA] = [A-].

$$K_a = \frac{[H^+][A^-]}{[HA]} = [H^+], \text{ since } \frac{[A^-]}{[HA]} = 1$$

- Since $K_a = [H^+]$, pH = pKa at the half-equivalence point.
- **Limitation**: This method becomes less accurate for very dilute acid solutions because water autoionization becomes significant.

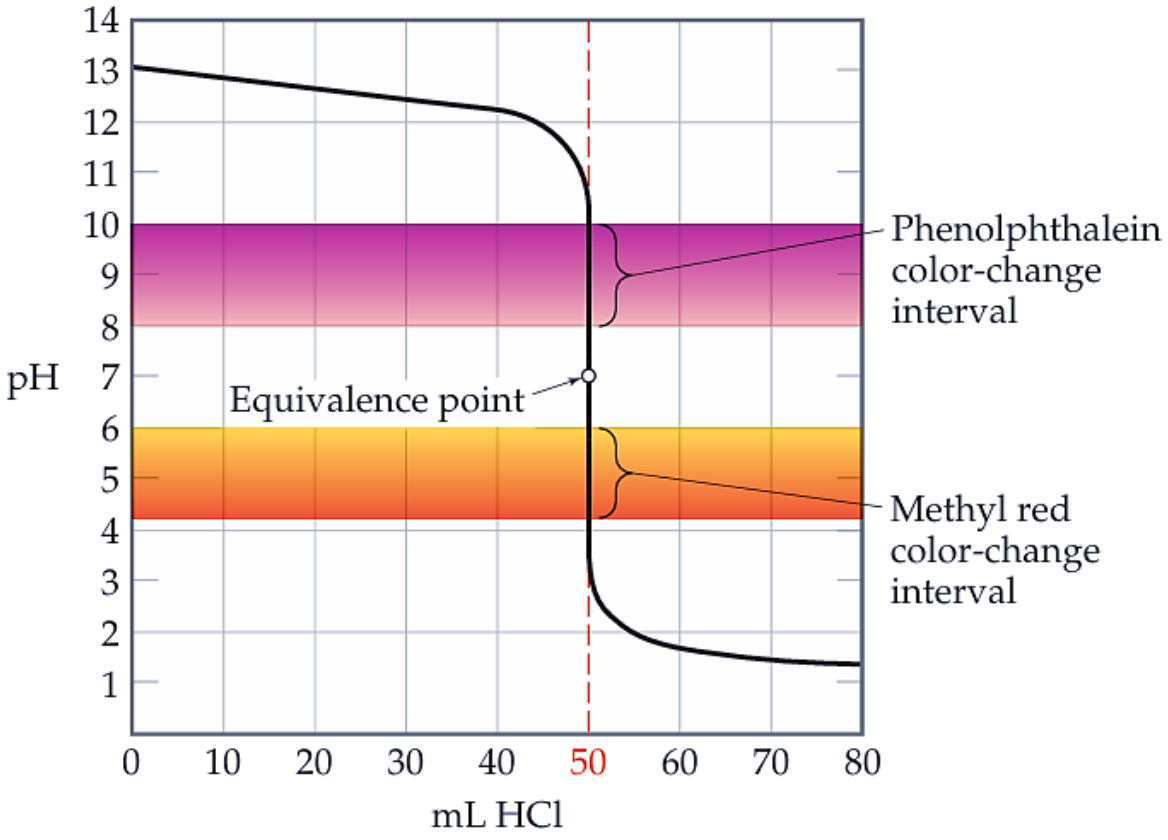
Acid-Base Titrations

Titrating with an Acid–Base Indicator

Q. How do you monitor pH change during titration?

- ✓ Using an indicator
- ✓ Using a pH meter

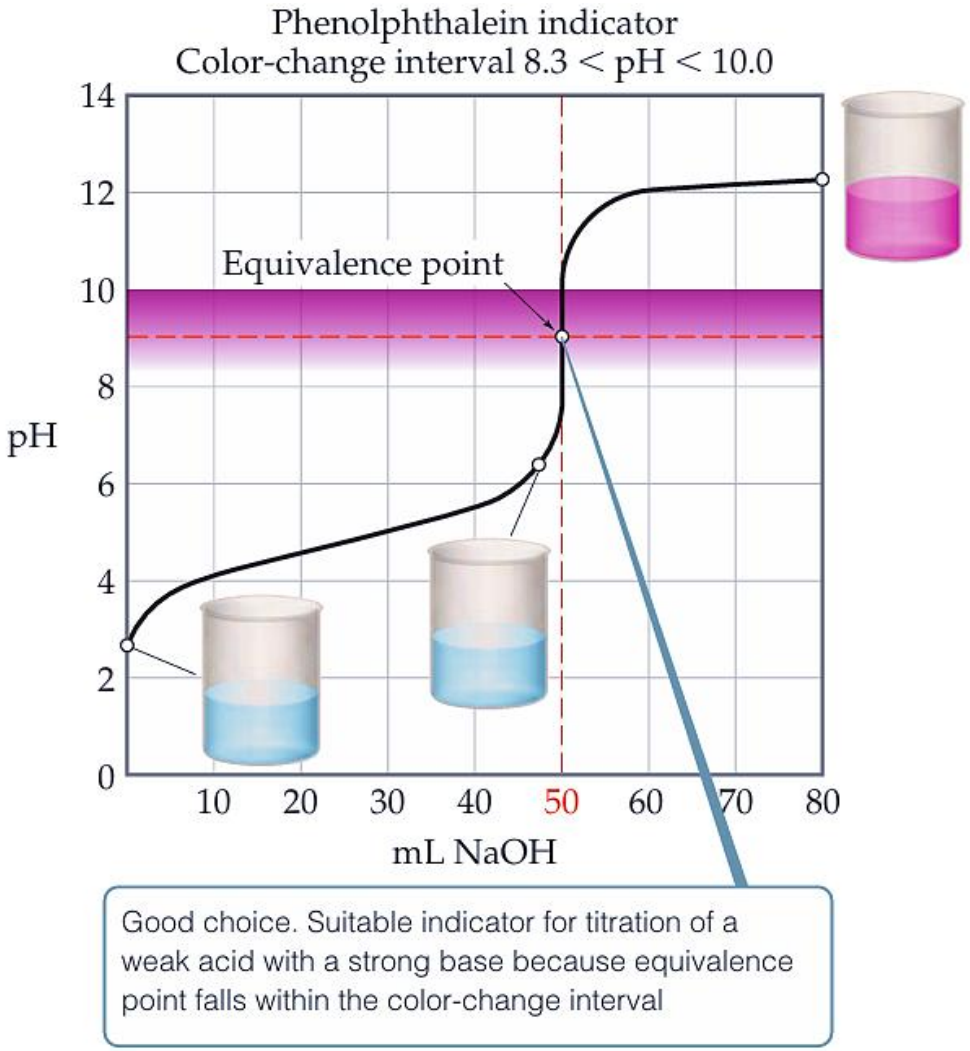
- An ideal indicator should change color at or near the equivalence point.
- Near the equivalence point, a single drop of titrant can cause a large pH change (Rapid pH change).
- The end point is the indicator's color change, which closely approximates the equivalence point.

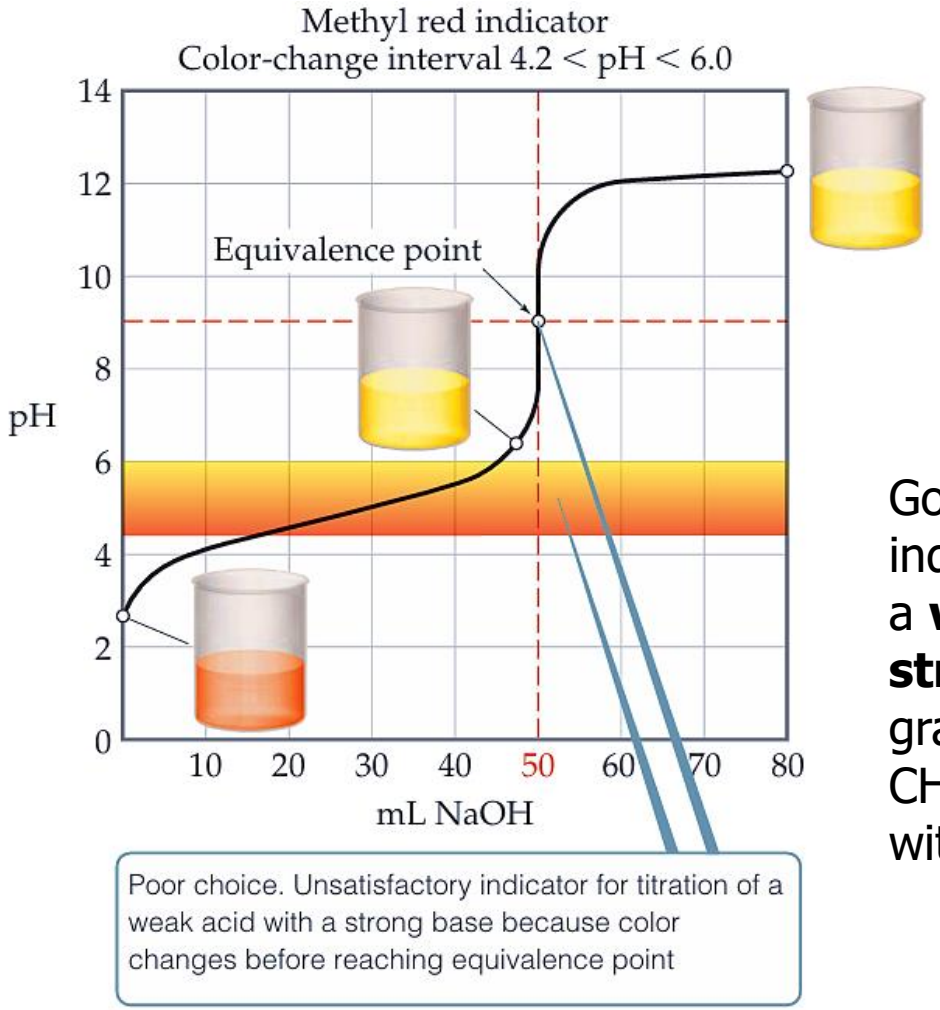


Strong base (NaOH) - Strong acid (HCl) titration

Acid-Base Titrations

Titrating with an Acid–Base Indicator

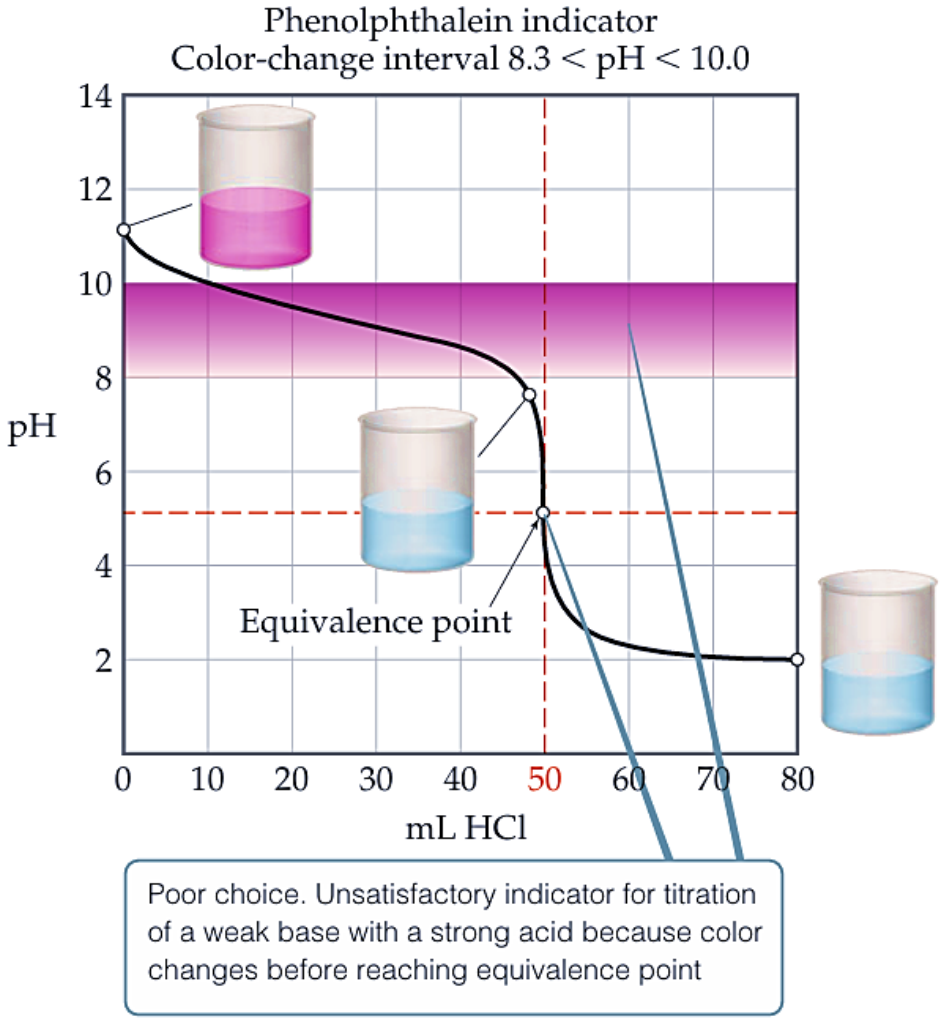


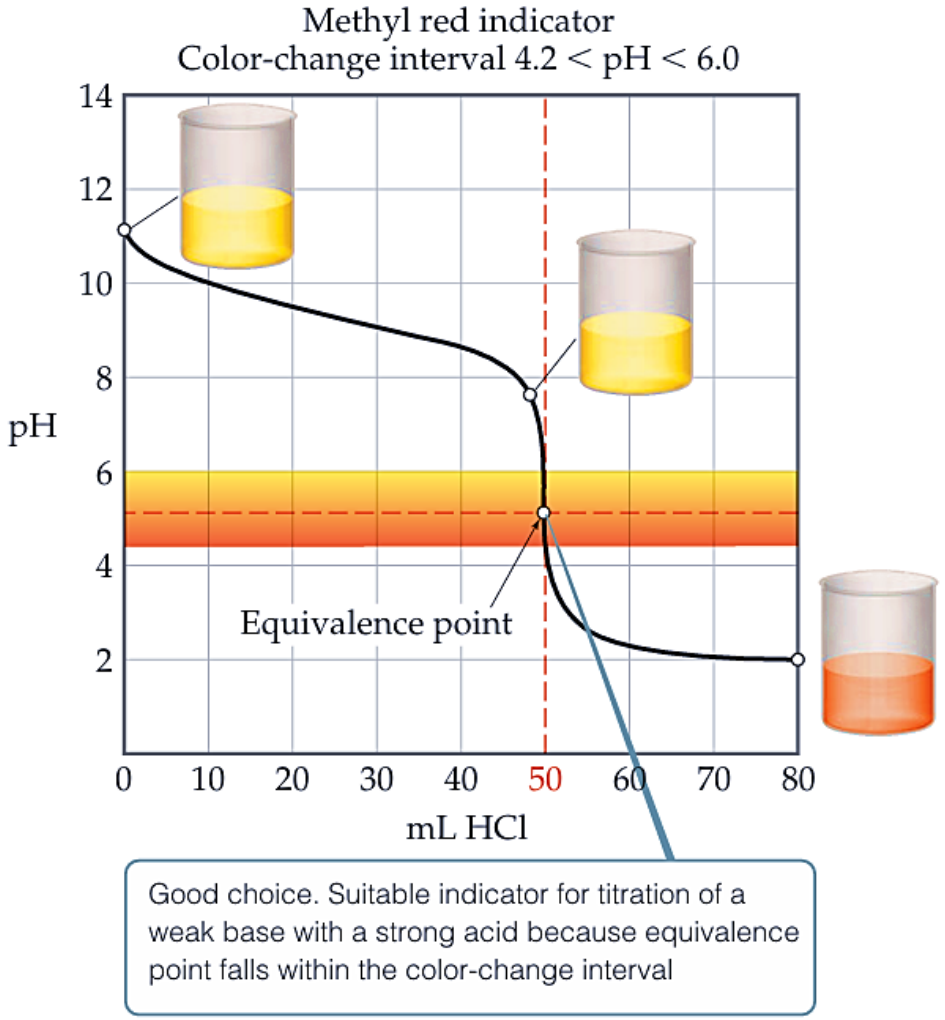


Good and poor indicators for titration of a **weak acid with a strong base** (In these graphs: 0.100 M CH_3COOH is titrated with 0.100 M NaOH)

Acid-Base Titrations

Titrating with an Acid–Base Indicator

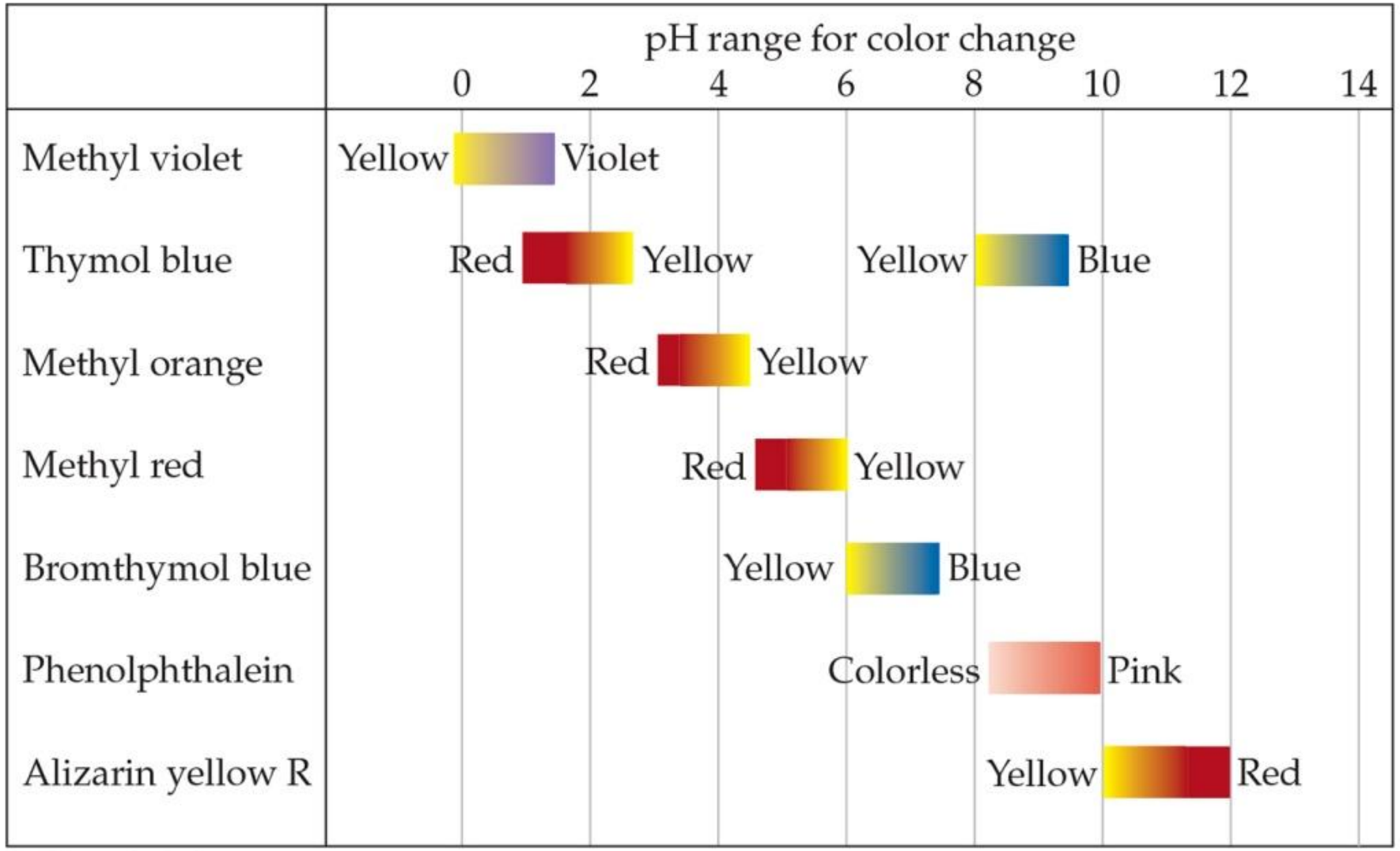




Good and poor indicators for titration of a **weak base with a strong acid** (In these graphs: 0.100 M NH3 is titrated with 0.100 M HCl).

Acid-Base Titrations

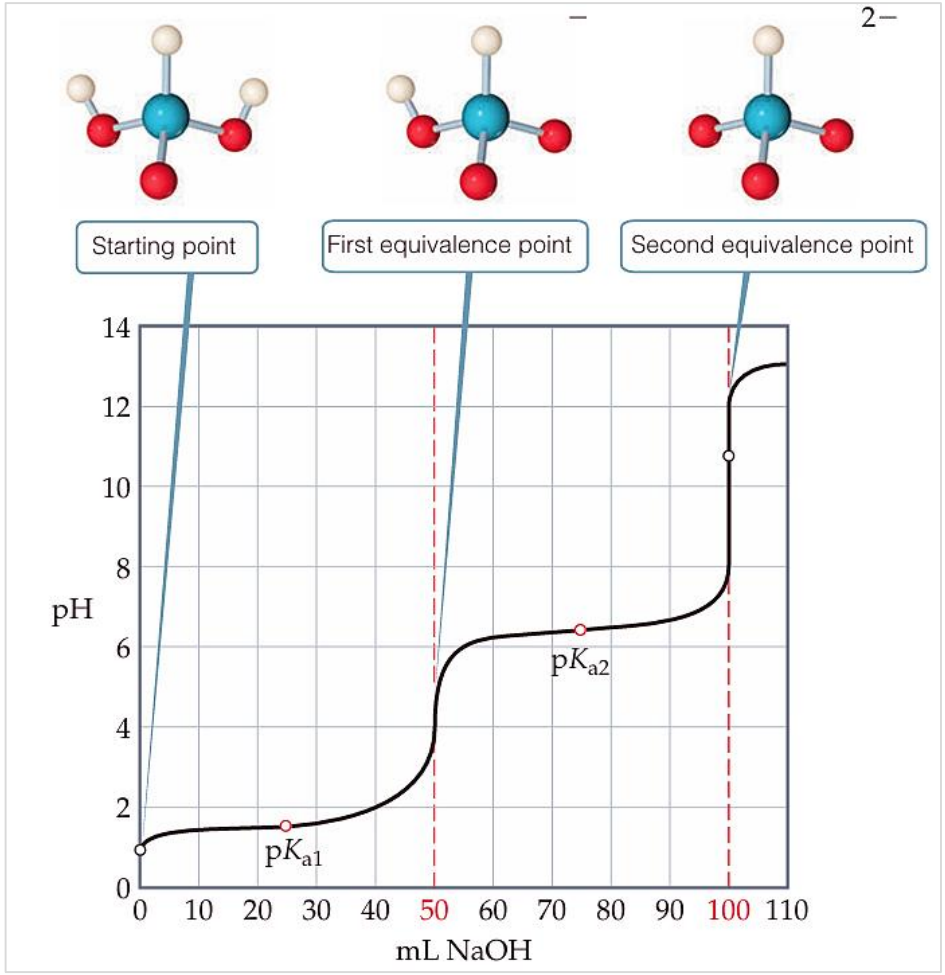
pH ranges for common acid–base indicators



Source: Chapter 16

Acid-Base Titrations

Titrations of Polyprotic Acids



Titration curve for a **diprotic acid**. The curve shows the pH change when 50.0 mL of 0.10 M H_3PO_3 is titrated with 0.10 M NaOH.

- Polyprotic weak acids are neutralized by OH^- in successive steps, with each ionizable H^+ removed separately (Example: H_3PO_3).
- Stepwise neutralization reactions

$$H_3PO_3\ (aq) + OH^-\ (aq) \rightarrow H_2PO_3^-\ (aq) + H_2O\ (l)$$

$$H_2PO_3^-\ (aq) + OH^-\ (aq) \rightarrow HPO_3^{2-}\ (aq) + H_2O\ (l)$$

- Acid dissociation equilibria and Henderson–Hasselbalch equations

$$H_3PO_3\ (aq) + OH^-\ (aq) \rightarrow H_2PO_3^-\ (aq) + H_2O\ (l)$$

$$K_{a1} = \frac{[H^+][H_2PO_3^-]}{[H_3PO_3]} \qquad \mathrm{pH} = \mathrm{p}K_{a1} + \frac{[H_2PO_3^-]}{[H_3PO_3]}$$

$$H_2PO_3^-\ (aq) + OH^-\ (aq) \rightarrow HPO_3^{2-}\ (aq) + H_2O\ (l)$$

$$K_{a2} = \frac{[H^+][H_2PO_3^-]}{[H_3PO_3]} \qquad \mathrm{pH} = \mathrm{p}K_{a2} + \frac{[HPO_3^{2-}]}{[H_2PO_3^-]}$$

Acid-Base Titrations

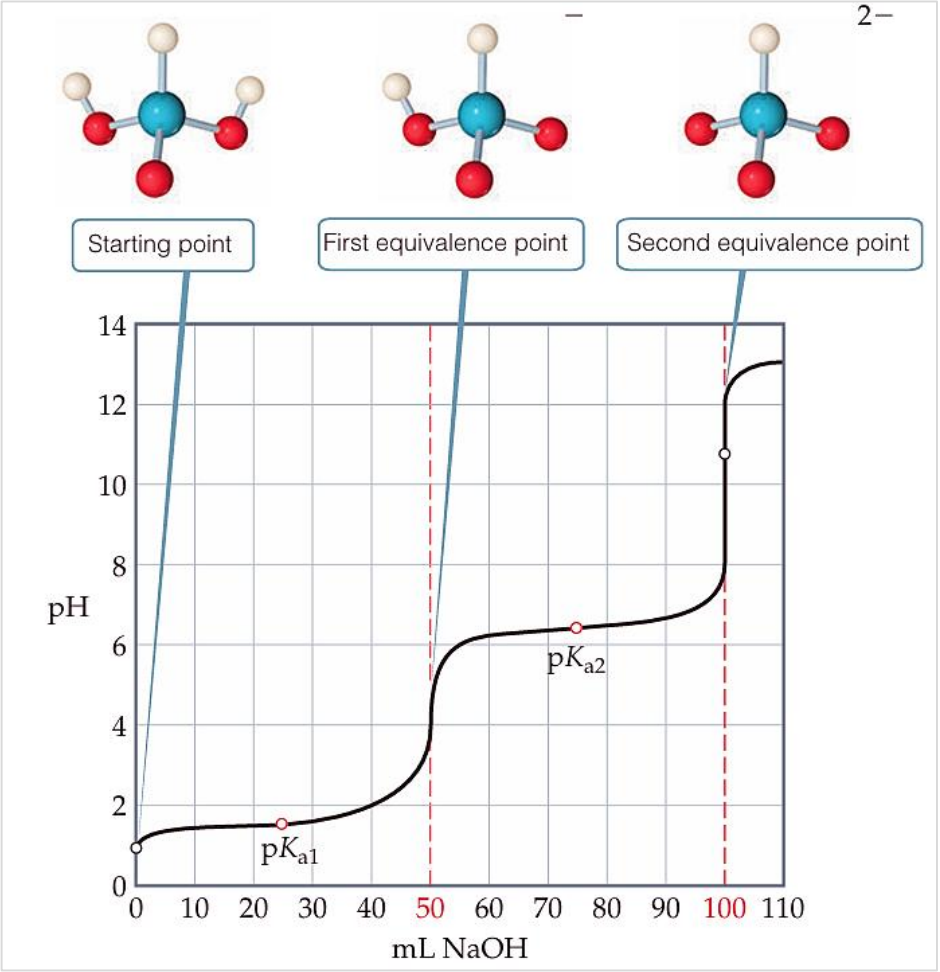
Q. The following curve shows the pH change when 50.0 mL of 0.10 M H_3PO_3 is titrated with 0.10 M NaOH.

a. What is the species present at the initial point? First equivalence point? Second equivalence point?

b. Calculate the pH at the initial state.

c. Calculate the pH at the first equivalence point (i.e. 50.0 mL NaOH added)

d. Calculate the pH at the second equivalence point (i.e. 100.0 mL NaOH added)



Titration curve for a **diprotic acid**. The curve shows the pH change when 50.0 mL of 0.10 M H_3PO_3 is titrated with 0.10 M NaOH.

Acid-Base Titrations

Q. A 20.0 mL sample of 0.150 M KOH is titrated with 0.125 M $HClO_4$ solution. Calculate the pH after the following volumes of **acid** have been added: (**a**) 20.0 mL, (**b**) 23.0 mL, (**c**) 24.0 mL, (**d**) 25.0 mL, (**e**) 30.0 mL.

Q. Consider the titration of 30.0 mL of 0.050 M NH_3 with 0.025 M HCl. Calculate the pH after the following volumes of **titrant** have been added: (**a**) 0 mL, (**b**) 20.0 mL, (**c**) 59.0 mL, (**d**) 60.0 mL, (**e**) 61.0 mL, (**f**) 65.0 mL.

Q. How many milliliters of 0.105 M HCl are needed to titrate each of the following solutions to the equivalence point: (**a**) 45.0 mL of 0.0950 M NaOH, (**b**) 22.5 mL of 0.118 M NH_3, (**c**) 125.0 mL of a solution that contains 1.35 g of NaOH per liter?

Q. Predict whether the equivalence point of each of the following titrations is below, above, or at pH 7: (**a**) benzoic acid titrated with KOH, (**b**) ammonia titrated with iodic acid, (**c**) hydroxylamine with hydrochloric acid.

Q. Calculate the pH at the equivalence point in titrating 0.100 M solutions of each of the following with 0.080 M NaOH: (**a**) hydrobromic acid, (**b**) chlorous acid, (**c**) benzoic acid.

Acid-Base Titrations

Q. At the half-equivalence point of a weak acid–strong base titration:

(a) pH = 7
(b) pH = pK_a
(c) [HA] = 0
(d) pH = 14 – pK_a

Answer: (b). Half of HA has been converted to A^-, so [HA] = [A^-] and pH = pK_a.

Q. NH_3 is titrated with HCl (equivalence point pH ≈ 5.3). Which indicator is best?

(a) Phenolphthalein (8.2–10.0)
(b) Thymol blue (8.0–9.6)
(c) Methyl red (4.2–6.3)
(d) Alizarin yellow R (10.1–12.0)

Answer: (c). Its colour-change range contains the equivalence-point pH.

Acid-Base Titrations

Summary points

- ✓ Equivalence point: moles of acid = moles of base (stoichiometry).
- ✓ Equivalence pH: SA–SB = 7; WA–SB > 7; WB–SA < 7. (W: weak, S: strong, A: Acid, B: Base)
- ✓ Before equivalence (weak acid): buffer region → Henderson–Hasselbalch; at half-equivalence pH = pK_a.
- ✓ After equivalence: the excess strong titrant controls the pH.
- ✓ Choose an indicator in which its colour change lies on the steep part of the curve.

Q. 25.0 mL of 0.100 M HCl is titrated with 0.100 M NaOH. What is the pH after 30.0 mL of NaOH has been added?

Answer: Excess OH^- = 3.00 – 2.50 = 0.50 mmol in 55.0 mL → $[OH^-] = 9.1 \times 10^{-3}$ M → pOH = 2.04 → **pH = 11.96**.

Solubility Equilibria

Q. Ba^{2+} ions are toxic, yet patients drink a $BaSO_4$ "barium meal" before an X-ray of the digestive tract. How can this be safe?

Answer: $BaSO_4$ is very sparingly soluble ($K_{sp} = 1.1 \times 10^{-10}$), so $[Ba^{2+}]$ in solution stays extremely low ($\approx 1 \times 10^{-5}$ M).

Q. What is solubility? Saturated solution?

Q. What is a precipitation reaction?

Solubility Equilibria

- A saturated solution exists in equilibrium with undissolved solute.
- The equilibrium constant for the dissolution of a sparingly soluble ionic solid is called the solubility product (K_{sp}).
- The magnitude of K_{sp} indicates the extent of dissolution at **a given temperature**.

"The solubility product K_{sp} of a compound equals the product of the concentrations of the ions involved in the equilibrium, each raised to the power of its **coefficient** in the equilibrium equation."

$$\mathbf{BaSO_4(s) \rightleftharpoons Ba^{2+}(aq) + SO_4^{2-}(aq)}$$

$$K_{sp} = [Ba^{2+}]\,[SO_4^{2-}]$$

$$K_{sp} = 1.1 \times 10^{-10} \text{ at } 25\ ^oC$$

$$\mathbf{Ba_3(PO_4)_2\,(s) \rightleftharpoons 3Ba^{2+}(aq) + 2PO_4^{3-}(aq)}$$

$$K_{sp} = [Ba^{2+}]^3\,[PO_4^{3-}]^2$$

Appendix D: List of K_{sp} at 25 ^oC

Solubility Equilibria

Solubility and K_{sp}

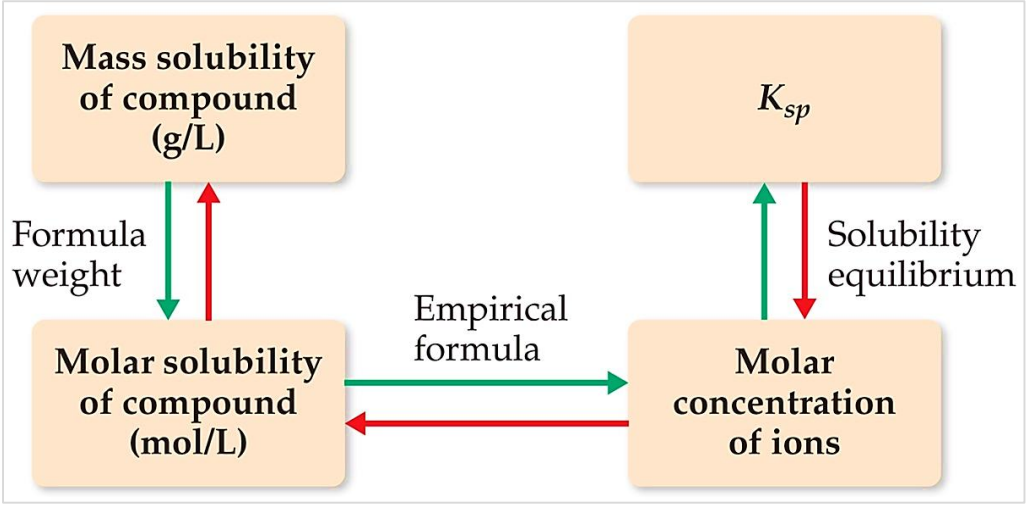
Q. How is solubility different from K_{sp}?

- **Solubility** refers to the amount of solute that dissolves to form a saturated solution at a given temperature, often expressed in **g/L**.
- **Molar solubility** is the number of moles of solute dissolved per liter of saturated solution (**mol/L**).
- **K_{sp}** is an equilibrium constant for the dissolution of a sparingly soluble ionic solid, and it is **dimensionless**.
- Although K_{sp} indicates the extent of dissolution, it cannot be directly compared with solubility without considering the dissolution stoichiometry.

Solubility Equilibria

Solubility and K_{sp}

- pH and the concentrations of other ions, especially common ions, can change solubility.
- The solubility of salts such as $Mg(OH)_2$ depends on solution pH.
- Adding a common ion generally decreases the solubility of a sparingly soluble salt.
- For a given salt, $\mathbf{K_{sp}}$ has a fixed value at a specific temperature.
- Solubility calculated from $\mathbf{K_{sp}}$ agrees best with experimental values for salts with low-charge ions that do not react with water.



Procedure for converting between solubility and K*sp*.

Solubility Equilibria

Representative Solubility Product Constants (K_{sp}) at 25 °C

Salt Name	Formula	Ion Dissolution Reaction	K_{sp} (at 25 °C)
Barium sulfate	$BaSO_4$	$BaSO_4(s) \rightleftharpoons Ba^{2+}(aq) + SO_4^{2-}(aq)$	1.1×10^{-10}
Calcium carbonate	$CaCO_3$	$CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$	4.5×10^{-9}
Calcium hydroxide	$Ca(OH)_2$	$Ca(OH)_2(s) \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq)$	6.5×10^{-6}
Silver chloride	$AgCl$	$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$	1.8×10^{-10}
Lead(II) sulfate	$PbSO_4$	$PbSO_4(s) \rightleftharpoons Pb^{2+}(aq) + SO_4^{2-}(aq)$	6.3×10^{-7}
Copper(II) carbonate	$CuCO_3$	$CuCO_3(s) \rightleftharpoons Cu^{2+}(aq) + CO_3^{2-}(aq)$	2.3×10^{-10}
Lead(II) chromate	$PbCrO_4$	$PbCrO_4(s) \rightleftharpoons Pb^{2+}(aq) + CrO_4^{2-}(aq)$	2.8×10^{-13}

Appendix D: Full List of **K_{sp} at 25 °C**

Q. Identify the least soluble salt.

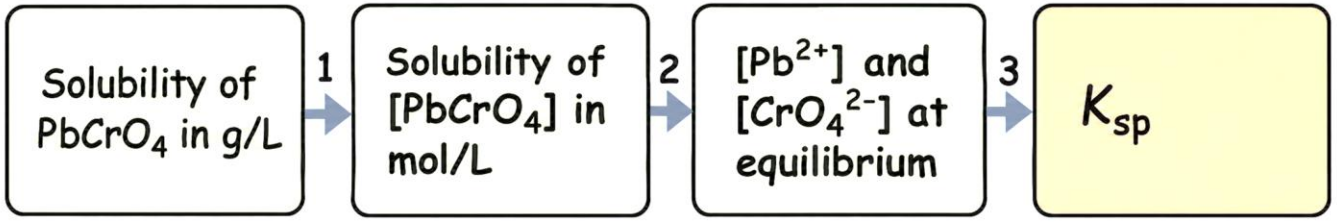
Q. Identify the most soluble salt.

Q. What are the possible reasons that caused these solubility differences? Explain

Solubility Equilibria

Q. The solubility of $PbCrO_4$ is 4.6×10^{-6} g/L. Determine the solubility product equilibrium constant for $PbCrO_4$.

$$PbCrO_4(s) \rightleftharpoons Pb^{2+}(aq) + CrO_4^{2-}(aq)$$



Step 1 conversion (using the molar mass of $PbCrO_4$)

$$(4.6 \times 10^{-6}\ \textbf{g/L}\ PbCrO_4) \times \left(\frac{1\ mol\ PbCrO_4}{323.2\ g\ PbCrO_4}\right) = 1.4 \times 10^{-8}\ \textbf{M}\ PbCrO_4$$

Step 2 conversion (using Stoichiometry):

1 mol of $PbCrO_4$ gives 1 mol of $Pb^{2+}(aq)$ and 1 mol of $CrO_4^{2-}(aq)$

$[Pb^{2+}] = [CrO_4^{2-}] = 1.4 \times 10^{-8}$ M

Step 3(K_{sp} calculation): $\boldsymbol{K_{sp} = [Pb^{2+}][CrO_4^{2-}]} = (1.4 \times 10^{-8})(1.4 \times 10^{-8}) = \mathbf{2.0 \times 10^{-16}}$

Solubility Equilibria

Q. At 25 °C, a saturated Ag_2CrO_4 solution has $[Ag^+] = 1.3 \times 10^{-4}$ M. Calculate K_{sp} for Ag_2CrO_4. Assume no other significant equilibria occur.

$$Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq) \qquad K_{sp} = [Ag^+]^2\,[CrO_4^{2-}]$$

Given: $[Ag^+] = 1.3 \times 10^{-4}$ M; $[CrO_4^{2-}] = ?$

Stoichiometry: for each CrO_4^{2-} ion there are two Ag+ ions in solution.

$$[CrO_4^{2-}] = (1.3 \times 10^{-4} \text{ mol } Ag^+/L) \times (1 \text{ mol } CrO_4^{2-} / 2 \text{ mol } Ag^+) = 6.5 \times 10^{-5} \text{ M}$$

$$K_{sp} = [Ag^+]^2\,[CrO_4^{2-}]$$

$$K_{sp} = (1.3 \times 10^{-4})^2(6.5 \times 10^{-5}) = \mathbf{1.1 \times 10^{-12}}$$

Q. At 298 K, a saturated $Cu_3(PO_4)_2$ solution has $[Cu^{2+}] = 5.01 \times 10^{-8}$ M. Calculate K_{sp} for $Cu_3(PO_4)_2$.

Solubility Equilibria

Q. The K_{sp} for CaF_2 is 3.9×10^{-11} at 25 °C. Assuming equilibrium is established between solid and dissolved CaF_2, and that there are no other important equilibria affecting its solubility, calculate the solubility of CaF_2 in grams per liter.

$$CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2\,F^-(aq)$$

	$CaF_2(s)$	$Ca^{2+}(aq)$	$F^-(aq)$
Initial concentration (M)	—	0	0
Change (M)	—	$+x$	$+2x$
Equilibrium concentration (M)	—	x	$2x$

1. Solubility product expression

$$K_{sp} = [Ca^{2+}][F^-]^2 = (x)(2x)^2 = 4x^3 = 3.9 \times 10^{-11}$$

2. Calculate the molar solubility

$$x = \sqrt[3]{\frac{3.9 \times 10^{-11}}{4}} = 2.1 \times 10^{-4}\ \text{mol L}^{-1}$$

3. Convert molar solubility to mass solubility

$$\frac{2.1 \times 10^{-4}\ \text{mol } CaF_2}{1\ \text{L solution}} \times \frac{78.1\ \text{g } CaF_2}{1\ \text{mol } CaF_2} = 1.6 \times 10^{-2}\ \text{g } CaF_2/\text{L solution}$$

Limitations of Solubility Products

- ✓ Ion interactions and ion pairing can cause calculated solubilities from K_{sp} to differ from experimental values. These effects increase with ion concentration and charge.
- ✓ Other equilibria and incomplete dissociation can increase solubility. For example, carbonate hydrolysis increases $CaCO_3$ solubility, and MgF_2 can form MgF^+ ion pairs in solution.

Solubility Equilibria

Q. Which is the correct K_{sp} expression for $Ca_3(PO_4)_2$?

(a) $[Ca^{2+}][PO_4^{3-}]$
(b) $[Ca^{2+}]^3[PO_4^{3-}]^2$
(c) $3[Ca^{2+}] \times 2[PO_4^{3-}]$
(d) $[Ca^{2+}]^3[PO_4^{3-}]^2 / [Ca_3(PO_4)_2]$

Answer: (b). Each ion is raised to its coefficient; the solid is not included.

Q. $AgCl$ ($K_{sp} = 1.8 \times 10^{-10}$) or Ag_2CrO_4 ($K_{sp} = 1.1 \times 10^{-12}$): which has the higher molar solubility?

(a) $AgCl$
(b) Ag_2CrO_4
(c) They are equal
(d) Cannot be determined

Answer: (b). $AgCl$: $s = \sqrt{K_{sp}} = 1.3 \times 10^{-5}$ M; Ag_2CrO_4: $4s^3 = K_{sp}$, $s = 6.5 \times 10^{-5}$ M.

Solubility Equilibria

Summary points

- ✓ K_{sp} = product of the ion concentrations, each raised to its coefficient; solids are omitted.
- ✓ Solubility (g/L) ↔ molar solubility (mol/L) ↔ ion concentrations ↔ K_{sp}.
- ✓ MX: $K_{sp} = s^2$; MX_2: $K_{sp} = 4s^3$; M_3X_2: $K_{sp} = 108s^5$.
- ✓ Compare K_{sp} values directly only for salts with the same ion ratio.
- ✓ Ion pairing and side equilibria can make real solubility larger than calculated.

Q. $Mg(OH)_2$ has $K_{sp} = 1.8 \times 10^{-11}$. Calculate its molar solubility.

Answer: $K_{sp} = [Mg^{2+}][OH^-]^2 = (s)(2s)^2 = 4s^3 \rightarrow s = 1.7 \times 10^{-4}$ M.

Factors That Affect Solubility

Q. Limestone caves form when rainwater containing dissolved CO_2 (a weak acid) slowly dissolves $CaCO_3$ rock. Why does acid dissolve $CaCO_3$ but not $AgCl$?

Answer: CO_3^{2-} is a basic anion: H^+ removes it, so $CaCO_3 \rightleftharpoons Ca^{2+} + CO_3^{2-}$ shifts right. Cl^- (from a strong acid) does not react with H^+.

Q. List the factors that affect solubility.

(1) Presence of common ions

(2) Solution pH

(3) Presence of complexing agents.

Factors That Affect Solubility

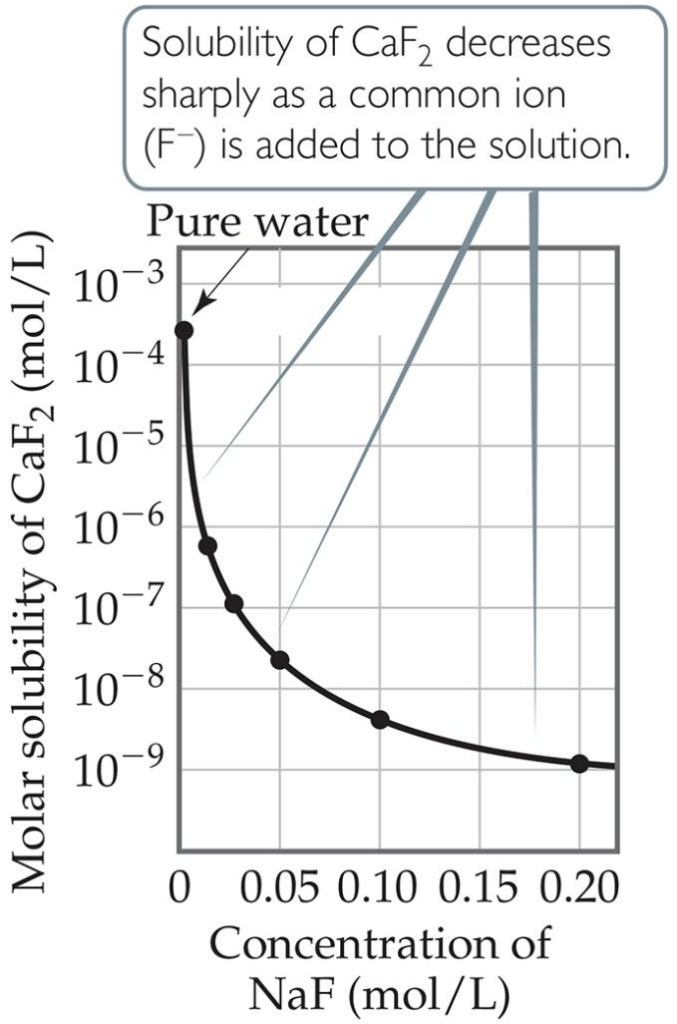
The Common-Ion Effect

Example:

$$CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2\,F^-(aq)$$

Addition of Ca^{2+} or F^- shifts equilibrium concentrations, reducing solubility

REVERSE SHIFT (COMMON ION EFFECT)
- Formation of precipitate [$CaF_2(s)$]
- Decreased ion concentrations in solution



Common-ion effect.

Factors That Affect Solubility

The Common-Ion Effect

Q. Calculate the **molar solubility** of CaF_2 (K_{sp}= 3.9 x 10^{-11}) at 25 °C in the presence of: **a.** 0.010 M in $Ca(NO_3)_2$ **b.** 0.010 M in NaF.

a. $CaF_2(s) \rightleftharpoons Ca^{2+}(aq) + 2F^-(aq)$

	$CaF_2(s)$	$Ca^{2+}(aq)$	$F^-(aq)$
Initial concentration (M)	—	0.010	0
Change (M)	—	+x	+2x
Equilibrium concentration (M)	—	(0.010 + x)	2x

$$K_{sp} = [Ca^{2+}][F^-]^2$$

$$3.9\ x\ 10^{-11} = (0.010 + x)\ (2x)^2 = (0.010)\ (2x)^2$$

$$\mathbf{x^2} = \frac{(3.9\ x\ 10^{-11})}{\mathbf{4(0.010)}}$$

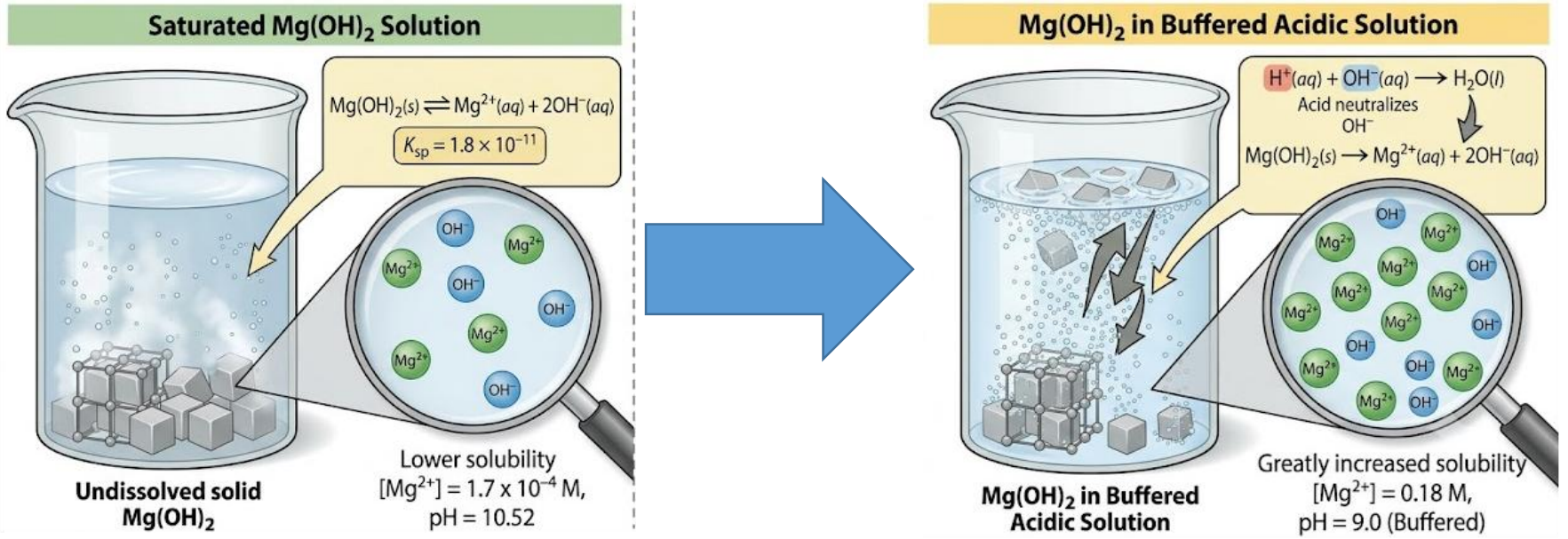
$$x = \sqrt{9.8 \times 10^{-10}} = 3.1 \times 10^{-5}\ M$$

b. 3.9 x 10^{-7} M

Factors That Affect Solubility

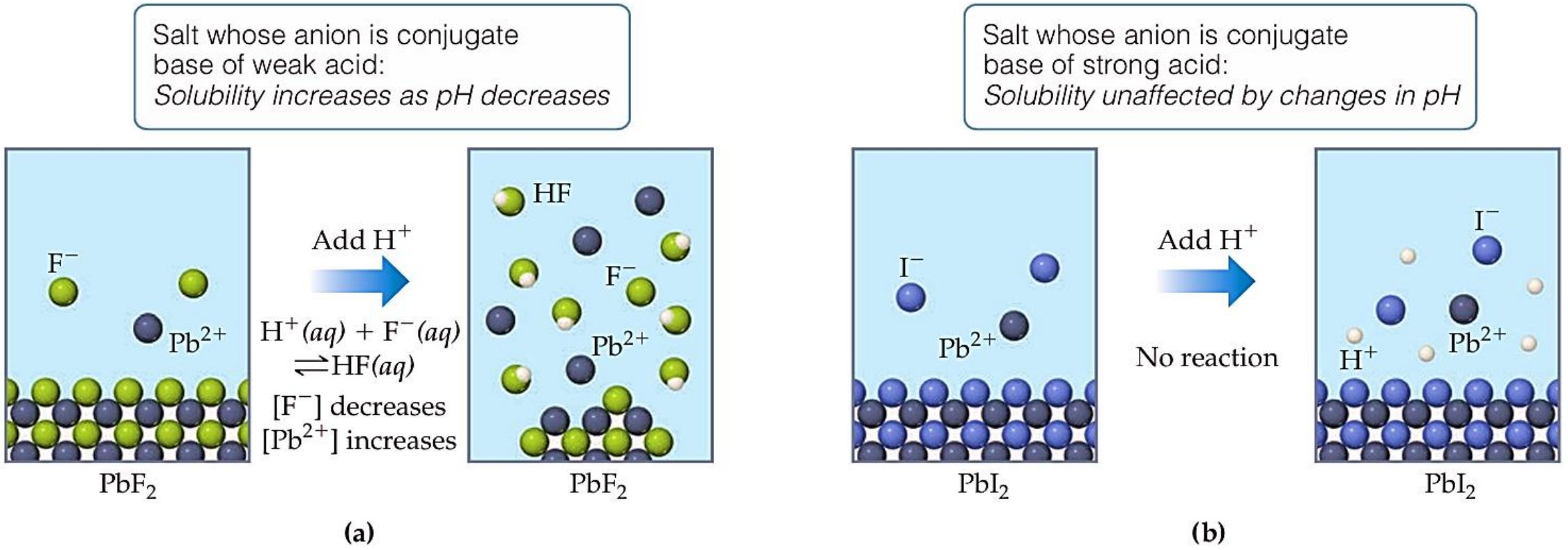
Solubility and pH

Compound Type	Example	pH Effect on Solubility
Metal hydroxides containing OH^-	$Mg(OH)_2$	Solubility increases as pH decreases (i.e as $[H^+]$ increases)
Salts containing basic anions (conjugate bases of weak acids)	PbF_2; CO_3^{2-}; PO_4^{3-}; CN-; S^{2-}	
Salts containing anions of strong acids	Salts containing Cl^-; Br^-; I^-; NO_3^-	Solubility is generally unaffected by pH



Factors That Affect Solubility

Solubility and pH



Response of two ionic compounds to addition of a strong acid. (a) The solubility of PbF_2 increases upon addition of acid. (b) The solubility of Pbl2 is not affected by the addition of acid. The water molecules and the anion of the strong acid have been omitted for clarity.

Factors That Affect Solubility

Solubility and pH

Q. Which of these substances are more soluble in acidic solution than in basic solution: (a) $Ni(OH)_2(s)$, (b) $CaCO_3(s)$, (c) $BaF_2(s)$, (d) $AgCl(s)$?

Solve

(a) $Ni(OH)_2(s)$ is more soluble in acidic solution because of the basicity of OH^-; the H^+ reacts with the OH^- ion, forming water:

$$Ni(OH)_2(s) \rightleftharpoons Ni^{2+}(aq) + 2\ OH^-(aq)$$
$$2\ OH^-(aq) + 2\ H^+(aq) \longrightarrow 2\ H_2O(l)$$
$$\text{Overall: } Ni(OH)_2(s) + 2\ H^+(aq) \rightleftharpoons Ni^{2+}(aq) + 2\ H_2O(l)$$

(b) Similarly, $CaCO_3(s)$ dissolves in acid solutions because CO_3^{2-} is a basic anion:

The reaction between CO_3^{2-} and H^+ occurs in steps, with HCO_3^- forming first and H_2CO_3 forming in appreciable amounts only when $[H^+]$ is sufficiently high.

$$CaCO_3(s) \rightleftharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$$
$$CO_3^{2-}(aq) + 2\ H^+(aq) \rightleftharpoons H_2CO_3(aq)$$
$$H_2CO_3(aq) \rightleftharpoons CO_2(g) + H_2O(l)$$
$$\text{Overall: } CaCO_3(s) + 2\ H^+(aq) \rightleftharpoons Ca^{2+}(aq) + CO_2(g) + H_2O(l)$$

(c) The solubility of BaF_2 is enhanced by lowering the pH because F^- is a basic anion:

$$BaF_2(s) \rightleftharpoons Ba^{2+}(aq) + 2\ F^-(aq)$$
$$2\ F^-(aq) + 2\ H^+(aq) \rightleftharpoons 2\ HF(aq)$$
$$\text{Overall: } BaF_2(s) + 2\ H^+(aq) \rightleftharpoons Ba^{2+}(aq) + 2\ HF(aq)$$

(d) The solubility of AgCl is unaffected by changes in pH because Cl^- is the anion of a strong acid and therefore has negligible basicity.

Factors That Affect Solubility

Solubility and pH

Tooth Decay and Fluoridation

1. Tooth decay: enamel dissolves

- Enamel is mainly hydroxyapatite, $Ca_{10}(PO_4)_6(OH)_2$.
- Bacteria in plaque break down sugars and produce acids (H^+).
- H^+ reacts with the basic OH^- and PO_4^{3-} ions, so the enamel dissolves and is washed away by saliva → cavities.

$$Ca_{10}(PO_4)_6(OH)_2(s) + 8\,H^+(aq) \longrightarrow 10\,Ca^{2+}(aq) + 6\,HPO_4^{2-}(aq) + 2\,H_2O(l)$$

Key facts

- Municipal water: ≈ 1 mg/L fluoride (1 ppm).
- Toothpaste: ≈ 0.1% fluoride by mass.
- Common sources: NaF, SnF_2, Na_2PO_3F, Na_2SiF_6.

2. Fluoridation: strengthening enamel

- F^- replaces OH^-, forming fluoroapatite, $Ca_{10}(PO_4)_6F_2$.
- F^- is a much weaker base than OH^-, so fluoroapatite resists acid attack.

Q. Why does fluoride make enamel less soluble in acid? (Think: which anion is the stronger base?)

Factors That Affect Solubility

Formation of Complex Ions

Q. Why do metal ions act as Lewis acids?

- A **Lewis acid** accepts an electron pair; a **Lewis base** donates an electron pair.
- Metal ions are positively charged and have empty valence orbitals, so they accept electron pairs.
- The electron-pair donors bound to a metal ion are **ligands**: H_2O, NH_3, CN^-, OH^-, Cl^-.
- Metal ion + ligands = **complex ion**, e.g. $Ag(NH_3)_2^+$, $Cu(NH_3)_4^{2+}$.

$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq)$

Lewis acid: Ag^+ Lewis base: NH_3

Formation constant

$K_f = [Ag(NH_3)_2^+] / ([Ag^+][NH_3]^2)$

$K_f = 1.7 \times 10^7$ at 25 °C

Q. What does a very large K_f tell you about the concentration of the free metal ion?

Answer: Almost all of the metal ion is bound in the complex, so the free $[M^{n+}]$ is very small.

Factors That Affect Solubility

Formation Constants, K_f

- K_f measures how strongly a metal ion binds its ligands.
- The larger K_f, the more stable the complex ion.
- Forming a complex removes free M^{n+} from solution, pulling the dissolution equilibrium to the right → **solubility increases**.

Q. Using Table 17.1, which silver complex is more stable: $Ag(NH_3)_2^+$ or $Ag(CN)_2^-$?

Answer: $Ag(CN)_2^-$, because $K_f = 1 \times 10^{21} \gg 1.7 \times 10^7$.

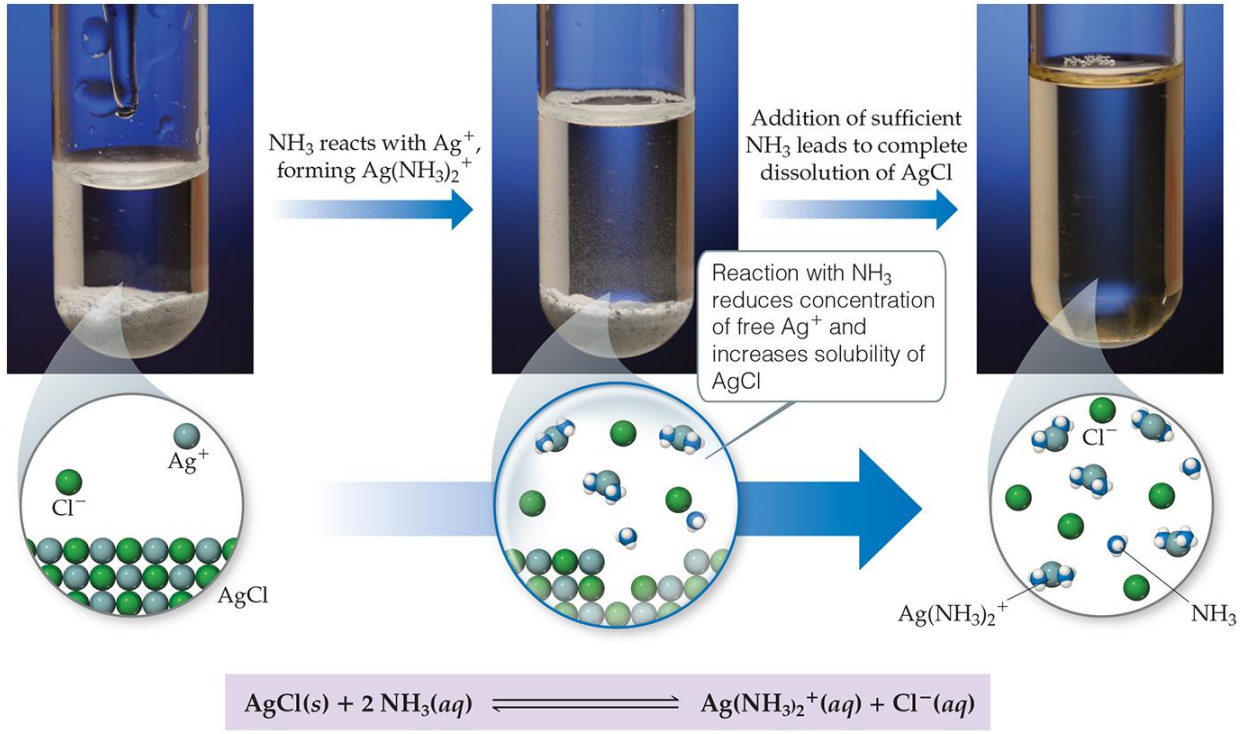
TABLE 17.1 Formation Constants for Some Metal Complex Ions in Water at 25 °C

Complex Ion	K_f	Chemical Equation
$Ag(NH_3)_2^+$	1.7×10^7	$Ag^+(aq) + 2\,NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq)$
$Ag(CN)_2^-$	1×10^{21}	$Ag^+(aq) + 2\,CN^-(aq) \rightleftharpoons Ag(CN)_2^-(aq)$
$Ag(S_2O_3)_2^{3-}$	2.9×10^{13}	$Ag^+(aq) + 2\,S_2O_3^{2-}(aq) \rightleftharpoons Ag(S_2O_3)_2^{3-}(aq)$
$Al(OH)_4^-$	1.1×10^{33}	$Al^{3+}(aq) + 4\,OH^-(aq) \rightleftharpoons Al(OH)_4^-(aq)$
$CdBr_4^{2-}$	5×10^3	$Cd^{2+}(aq) + 4\,Br^-(aq) \rightleftharpoons CdBr_4^{2-}(aq)$
$Cr(OH)_4^-$	8×10^{29}	$Cr^{3+}(aq) + 4\,OH^-(aq) \rightleftharpoons Cr(OH)_4^-(aq)$
$Co(SCN)_4^{2-}$	1×10^3	$Co^{2+}(aq) + 4\,SCN^-(aq) \rightleftharpoons Co(SCN)_4^{2-}(aq)$
$Cu(NH_3)_4^{2+}$	5×10^{12}	$Cu^{2+}(aq) + 4\,NH_3(aq) \rightleftharpoons Cu(NH_3)_4^{2+}(aq)$
$Cu(CN)_4^{2-}$	1×10^{25}	$Cu^{2+}(aq) + 4\,CN^-(aq) \rightleftharpoons Cu(CN)_4^{2+}(aq)$
$Ni(NH_3)_6^{2+}$	1.2×10^9	$Ni^{2+}(aq) + 6\,NH_3(aq) \rightleftharpoons Ni(NH_3)_6^{2+}(aq)$
$Fe(CN)_6^{4-}$	1×10^{35}	$Fe^{2+}(aq) + 6\,CN^-(aq) \rightleftharpoons Fe(CN)_6^{4-}(aq)$
$Fe(CN)_6^{3-}$	1×10^{42}	$Fe^{3+}(aq) + 6\,CN^-(aq) \rightleftharpoons Fe(CN)_6^{3-}(aq)$
$Zn(OH)_4^{2-}$	4.6×10^{17}	$Zn^{2+}(aq) + 4\,OH^-(aq) \rightleftharpoons Zn(OH)_4^{2-}(aq)$

Factors That Affect Solubility

How Complex-Ion Formation Affects Solubility

Q. Predict: what happens to solid *AgCl* as $NH_3(aq)$ is added?



- AgCl is sparingly soluble: $K_{sp} = 1.8 \times 10^{-10}$.
- NH_3 binds Ag^+ as $Ag(NH_3)_2^+$, lowering $[Ag^+]$.
- Le Châtelier: more AgCl dissolves to replace the Ag^+.

$$AgCl(s) \rightleftharpoons Ag^+ + Cl^- \quad (K_{sp})$$

$$Ag^+ + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+ \quad (K_f)$$

$$\mathbf{AgCl(s) + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+ + Cl^-}$$

$$\mathbf{K = K_{sp} \times K_f}$$

Factors That Affect Solubility

Complex-Ion Formation and Solubility

Q. Calculate the molar solubility of $AgCl$ in 1.0 M $NH_3(aq)$ and compare it with that in pure water. ($K_{sp} = 1.8 \times 10^{-10}$; K_f of $Ag(NH_3)_2^+ = 1.7 \times 10^7$)

Step 1. Add the two equilibria: $AgCl(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq) + Cl^-(aq)$

$K = K_{sp} \times K_f = (1.8 \times 10^{-10}) \times (1.7 \times 10^7) = 3.1 \times 10^{-3}$

Step 2. ICE table

	NH_3	$Ag(NH_3)_2^+$	Cl^-
Initial (M)	1.0	0	0
Change (M)	-2x	+x	+x
Equilibrium (M)	1.0 – 2x	x	x

Step 3. $K = x^2 / (1.0 - 2x)^2 = 3.1 \times 10^{-3}$

Take the square root of both sides:

$x / (1.0 - 2x) = 0.055$

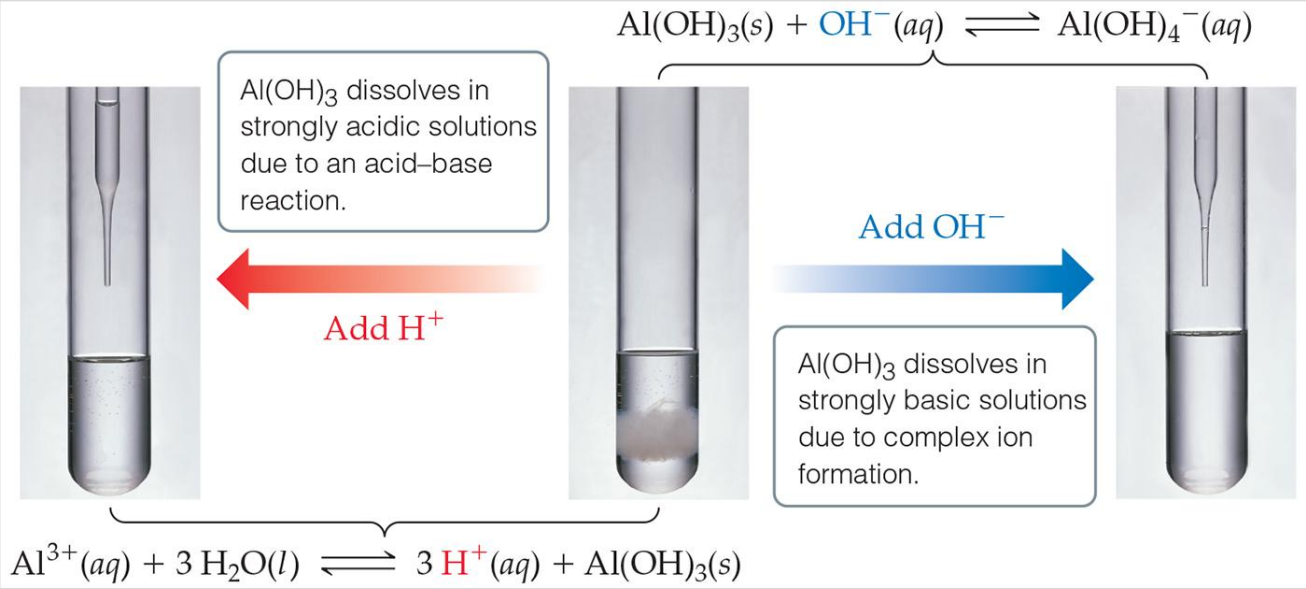
$x = 0.050$ M (molar solubility)

Step 4. In pure water: $s = \sqrt{K_{sp}} = 1.3 \times 10^{-5}$ M. $AgCl$ is $\approx$ **4×10^3 times** more soluble in 1.0 M NH_3!

Factors That Affect Solubility

Amphoterism and Solubility

Q. $Al(OH)_3$ is insoluble in water. Will it dissolve in strong acid? In strong base?



- **Amphoteric** oxides and hydroxides dissolve in both strong acids and strong bases.
- Examples: hydroxides of Al^{3+}, Cr^{3+}, Zn^{2+} and Sn^{2+}.

In acid: $Al(OH)_3(s) + 3H^+(aq) \rightarrow Al^{3+}(aq) + 3H_2O(l)$

In base: $Al(OH)_3(s) + OH^-(aq) \rightleftharpoons Al(OH)_4^-(aq)$

Application: excess NaOH dissolves $Al(OH)_3$ but not $Fe(OH)_3$, so Al^{3+} can be separated from Fe^{3+}.

Factors That Affect Solubility

Q. Calculate the molar solubility of AgBr in 1.0 M NH_3. ($K_{sp}(AgBr) = 5.0 \times 10^{-13}$; K_f of $Ag(NH_3)_2^+ = 1.7 \times 10^7$)

Answer: $K = K_{sp}K_f = 8.5 \times 10^{-6}$; $x / (1.0 - 2x) = 2.9 \times 10^{-3} \rightarrow x \approx \mathbf{2.9 \times 10^{-3}\ M}$

Q. Which of these is more soluble in 1.0 M NaOH than in pure water: $Zn(OH)_2$, $Mg(OH)_2$, $Al(OH)_3$?

Answer: $Zn(OH)_2$ and $Al(OH)_3$ (amphoteric: form $Zn(OH)_4^{2-}$ and $Al(OH)_4^-$). $Mg(OH)_2$ is **less** soluble (common ion OH^-).

Q. Which salt's solubility is NOT affected by pH: CaF_2, AgI, $Fe(OH)_3$, $BaCO_3$?

Answer: AgI. I^- is the anion of a strong acid (HI) and has negligible basicity.

Factors That Affect Solubility

Q. The molar solubility of $PbCl_2$ is lowest in:

(a) pure water
(b) 0.10 M $NaNO_3$
(c) 0.10 M NaCl
(d) 0.10 M HNO_3

Answer: (c). Cl^- is a common ion and shifts $PbCl_2(s) \rightleftharpoons Pb^{2+} + 2Cl^-$ to the left.

Q. Adding $NH_3(aq)$ to a suspension of solid AgCl:

(a) decreases its solubility
(b) increases its solubility
(c) has no effect
(d) forms more AgCl(s)

Answer: (b). $Ag(NH_3)_2^+$ forms, removing free Ag^+, so more AgCl dissolves.

Factors That Affect Solubility

Summary points

- ✓ Common ion → solubility decreases (equilibrium shifts left).
- ✓ Salts of basic anions (OH^-, F^-, CO_3^{2-}, PO_4^{3-}, S^{2-}) are more soluble in acid; salts of Cl^-, Br^-, I^-, NO_3^- are not affected by pH.
- ✓ Lewis-base ligands (NH_3, CN^-, OH^-) form complex ions with metal ions; overall $K = K_{sp} \times K_f$.
- ✓ Amphoteric hydroxides (Al^{3+}, Cr^{3+}, Zn^{2+}, Sn^{2+}) dissolve in both strong acid and strong base.

Q. Explain why $Al(OH)_3$ dissolves in both $HCl(aq)$ and $NaOH(aq)$, using one equation for each.

Answer: $Al(OH)_3 + 3H^+ \rightarrow Al^{3+} + 3H_2O$; $Al(OH)_3 + OH^- \rightleftharpoons Al(OH)_4^-$

Precipitation and Separation of Ions

Q. Mixing clear $AgNO_3(aq)$ and $NaCl(aq)$ instantly gives a white solid. Would a solid ALWAYS appear, no matter how dilute the two solutions are? Why or why not?

Answer: No. A precipitate forms only when the ion product Q exceeds K_{sp}; very dilute solutions can give $Q < K_{sp}$.

Precipitation and Separation of Ions

Will a Precipitate Form?

Q. When two solutions are mixed, how can we predict whether a precipitate will form?

- Calculate the ion product, **Q** (the reaction quotient), from the actual ion concentrations and compare it with $\mathbf{K_{sp}}$:

$Q < K_{sp}$	$Q = K_{sp}$	$Q > K_{sp}$
Unsaturated	**Saturated**	**Supersaturated**
More solid can dissolve – no precipitate forms.	The system is at equilibrium.	A precipitate forms until $Q = K_{sp}$.

- Use the ion concentrations **after mixing** (divide moles by the total volume!).
- Q has the same form as the K_{sp} expression, but uses non-equilibrium concentrations.

Precipitation and Separation of Ions

Q. Will a precipitate form when 0.10 L of 8.0×10^{-3} M $Pb(NO_3)_2$ is added to 0.40 L of 5.0×10^{-3} M Na_2SO_4? (K_{sp} of $PbSO_4 = 6.3 \times 10^{-7}$)

Step 1. Possible precipitate: $PbSO_4(s) \rightleftharpoons Pb^{2+}(aq) + SO_4^{2-}(aq)$ ($NaNO_3$ is soluble)

Step 2. Concentrations after mixing (V_{total} = 0.10 L + 0.40 L = 0.50 L):

$[Pb^{2+}] = (0.10\ L)(8.0 \times 10^{-3}\ M) / 0.50\ L = 1.6 \times 10^{-3}\ M$

$[SO_4^{2-}] = (0.40\ L)(5.0 \times 10^{-3}\ M) / 0.50\ L = 4.0 \times 10^{-3}\ M$

Step 3. $Q = [Pb^{2+}][SO_4^{2-}] = (1.6 \times 10^{-3})(4.0 \times 10^{-3}) = 6.4 \times 10^{-6}$

Step 4. $Q\ (6.4 \times 10^{-6}) > K_{sp}\ (6.3 \times 10^{-7}) \rightarrow$ **$PbSO_4$ precipitates.**

Precipitation and Separation of Ions

Q. Will a precipitate form when 0.050 L of 2.0×10^{-2} M NaF is mixed with 0.010 L of 1.0×10^{-2} M $Ca(NO_3)_2$? (K_{sp} of $CaF_2 = 3.9 \times 10^{-11}$)

Answer: $[Ca^{2+}] = 1.7 \times 10^{-3}$ M, $[F^-] = 1.7 \times 10^{-2}$ M; $Q = 4.6 \times 10^{-7} > K_{sp}$ → **yes, CaF_2 precipitates.**

Q. Will AgCl precipitate when 1.0 mL of 1.0×10^{-4} M NaCl is added to 1.0 L of 1.0×10^{-4} M $AgNO_3$? ($K_{sp} = 1.8 \times 10^{-10}$)

Answer: $[Cl^-] \approx 1.0 \times 10^{-7}$ M, $[Ag^+] \approx 1.0 \times 10^{-4}$ M; $Q = 1.0 \times 10^{-11} < K_{sp}$ → **no precipitate.**

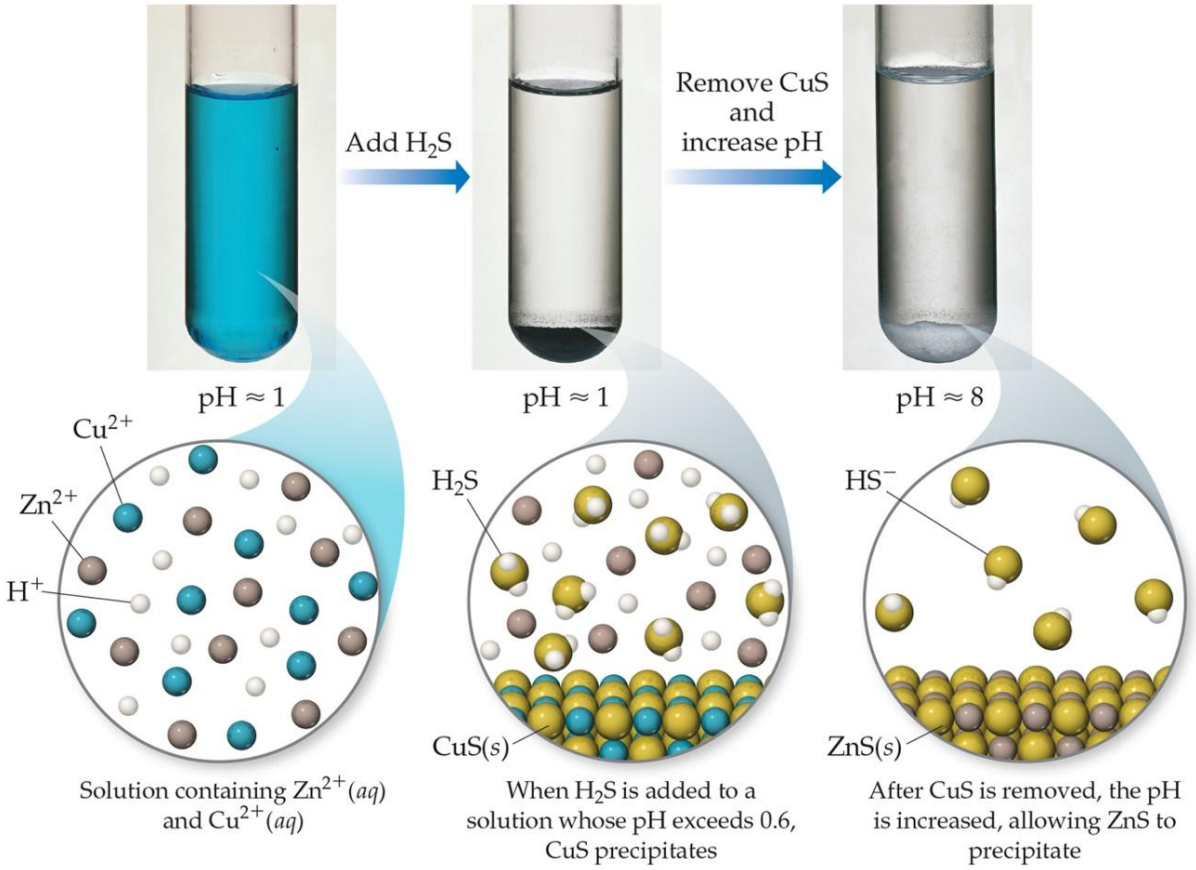
Q. What minimum $[OH^-]$ is needed to begin precipitating $Mg(OH)_2$ from 0.050 M Mg^{2+}? ($K_{sp} = 1.8 \times 10^{-11}$)

Answer: $[OH^-] = \sqrt{(K_{sp} / [Mg^{2+}])} = \sqrt{(3.6 \times 10^{-10})} = \mathbf{1.9 \times 10^{-5}\ M}$ (pH ≈ 9.28)

Precipitation and Separation of Ions

Selective Precipitation of Ions

Q. How can two metal ions dissolved in the same solution be separated?



1. Add a reagent (e.g. Cl^-, SO_4^{2-}, S^{2-}, OH^-) that forms a much less soluble salt with one of the ions.
2. The salt whose Q exceeds K_{sp} first (at the lowest reagent concentration) precipitates first.
3. Filter off the solid; the other ion stays in solution.

- *Figure:* at pH ≈ 1, H_2S precipitates CuS but not ZnS; after CuS is removed, raising the pH to ≈ 8 precipitates ZnS.

Precipitation and Separation of Ions

Selective Precipitation

Q. A solution contains 1.0×10^{-2} M Ag^+ and 2.0×10^{-2} M Pb^{2+}. Cl^- is added slowly. What $[Cl^-]$ is needed to begin precipitating AgCl ($K_{sp} = 1.8 \times 10^{-10}$) and $PbCl_2$ ($K_{sp} = 1.7 \times 10^{-5}$)? Which precipitates first?

Step 1. AgCl: $K_{sp} = [Ag^+][Cl^-] \rightarrow [Cl^-] = (1.8 \times 10^{-10}) / (1.0 \times 10^{-2}) = \mathbf{1.8 \times 10^{-8}\ M}$

Step 2. $PbCl_2$: $K_{sp} = [Pb^{2+}][Cl^-]^2 \rightarrow [Cl^-] = \sqrt{[(1.7 \times 10^{-5}) / (2.0 \times 10^{-2})]} = \mathbf{2.9 \times 10^{-2}\ M}$

Step 3. AgCl needs about 10^6 times less Cl^-, so **AgCl precipitates first**; Pb^{2+} stays in solution.

Q. When $PbCl_2$ just begins to precipitate, what $[Ag^+]$ is left in solution?

Answer: $[Ag^+] = K_{sp} / [Cl^-] = (1.8 \times 10^{-10}) / (2.9 \times 10^{-2}) = 6.2 \times 10^{-9}$ M → more than 99.99% of the Ag^+ has already precipitated.

Precipitation and Separation of Ions

Qualitative Analysis of Metal Ions

Q. How can we identify which metal ions are present in an unknown solution?

Group of cations by successive selective precipitation:

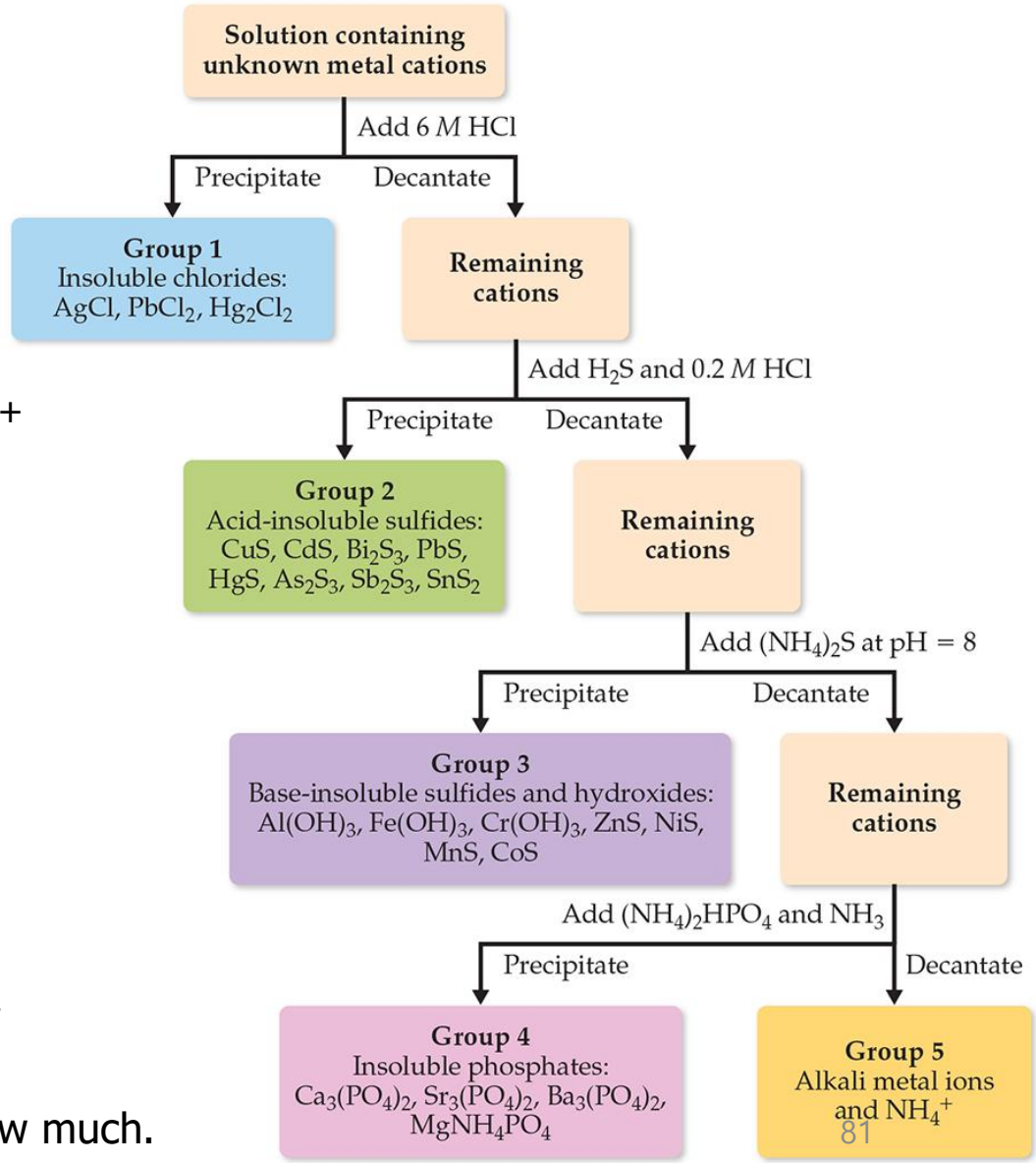
Group 1 – insoluble chlorides (6 M HCl): Ag^+, Pb^{2+}, Hg_2^{2+}

Group 2 – acid-insoluble sulfides (H_2S, 0.2 M HCl): Cu^{2+}, Cd^{2+}, Bi^{3+}, Hg^{2+}, …

Group 3 – base-insoluble sulfides and hydroxides ($(NH_4)_2S$, pH 8): Al^{3+}, Fe^{3+}, Cr^{3+}, Zn^{2+}, Ni^{2+}, …

Group 4 – insoluble phosphates ($(NH_4)_2HPO_4$, NH_3): Mg^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}

Group 5 – remain in solution: alkali metal ions and NH_4^+



Qualitative analysis tells us what is present; quantitative analysis tells us how much.

Precipitation and Separation of Ions

Q. For a mixture, $Q = 2.0 \times 10^{-8}$ and $K_{sp} = 1.0 \times 10^{-10}$. What happens?

(a) Unsaturated; no precipitate
(b) Saturated; at equilibrium
(c) A precipitate forms
(d) More solid dissolves

Answer: (c). $Q > K_{sp}$, so solid forms until $Q = K_{sp}$.

Q. In the qualitative-analysis scheme, which ion precipitates when 6 M HCl is added?

(a) Cu^{2+}
(b) Ag^{+}
(c) Zn^{2+}
(d) Na^{+}

Answer: (b). Group 1 (insoluble chlorides): $AgCl$, $PbCl_2$, Hg_2Cl_2.

Precipitation and Separation of Ions

Summary points

- ✓ $Q < K_{sp}$: unsaturated, no precipitate; $Q = K_{sp}$: saturated; $Q > K_{sp}$: precipitate forms.
- ✓ Always use the ion concentrations after mixing (total volume).
- ✓ The minimum concentration needed to start precipitation comes from solving K_{sp} for that ion.
- ✓ The salt that needs the lowest reagent concentration precipitates first (selective precipitation).
- ✓ Qualitative analysis separates cations into five groups by successive selective precipitation.

Q. A solution is 0.010 M in both Ba^{2+} and Sr^{2+}. SO_4^{2-} is added slowly. Which sulfate precipitates first? (K_{sp}: $BaSO_4 = 1.1 \times 10^{-10}$; $SrSO_4 = 3.2 \times 10^{-7}$)

Answer: $BaSO_4$ needs $[SO_4^{2-}] = 1.1 \times 10^{-8}$ M; $SrSO_4$ needs 3.2×10^{-5} M → **$BaSO_4$ precipitates first.**

Chapter 17 Summary

Topic	Key relationship
Common-ion effect	A common ion shifts the equilibrium to the left → less ionization, lower solubility
Buffer pH	$pH = pK_a + \log([base]/[acid])$; $pH = pK_a$ when [base] = [acid]
Strong acid/base added to a buffer	Stoichiometry first (moles), then Henderson–Hasselbalch
Titration curves	Equivalence pH: SA–SB = 7, WA–SB > 7, WB–SA < 7; half-equivalence: $pH = pK_a$
Solubility product	$M_mX_n(s) \rightleftharpoons mM^{n+}(aq) + nX^{m-}(aq)$; $K_{sp} = [M^{n+}]^m[X^{m-}]^n$
Factors affecting solubility	Common ion ↓; acid ↑ (basic anions); complex ions ↑ ($K = K_{sp} \times K_f$)
Precipitation	$Q > K_{sp}$: precipitate forms; $Q < K_{sp}$: no precipitate

Learning Outcomes

By now, you should be able to:

- ✓ Explain the common-ion effect and calculate pH when a common ion is present.
- ✓ Describe how buffers work and calculate buffer pH, including after adding strong acid or base.
- ✓ Interpret titration curves and choose a suitable indicator.
- ✓ Write K_{sp} expressions and convert between solubility and K_{sp}.
- ✓ Predict how common ions, pH and complex-ion formation change solubility.
- ✓ Use Q vs K_{sp} to predict precipitation and to separate ions.

End-of-Chapter Exercises

Practice with textbook exercises – Chapter 17

- The following **15 exercises** are taken from the end of Chapter 17 of *Chemistry: The Central Science* (15th ed., SI units). Each is labelled with its exercise number.
- They cover every section of the chapter and are ordered section by section.
- Try each problem first; the worked answer appears on the next click.
- Data (Appendix C, Appendix D, Tables) are those of the textbook.

Constant values for Chapter 17 Questions

Values from Appendix D of the textbook (25 °C), selected for the end-of-chapter exercises that follow.

Type	Substance	Formula	Value
K_a	Acetic acid	CH_3COOH	1.8×10^{-5}
K_a	Propanoic (propionic) acid	C_2H_5COOH	1.3×10^{-5}
K_a	Lactic acid	$CH_3CH(OH)COOH$	1.4×10^{-4}
K_a	Formic acid	$HCOOH$	1.8×10^{-4}
K_{a1}	Carbonic acid	H_2CO_3	4.3×10^{-7}
K_b	Trimethylamine	$(CH_3)_3N$	6.4×10^{-5}
K_w	Water	H_2O	1.0×10^{-14}
K_{sp}	Silver iodide	AgI	8.3×10^{-17}
K_{sp}	Calcium hydroxide	$Ca(OH)_2$	6.5×10^{-6}
K_{sp}	Silver sulfate	Ag_2SO_4	1.5×10^{-5}
K_{sp}	Calcium fluoride	CaF_2	3.9×10^{-11}
K_{sp}	Lanthanum fluoride	LaF_3	2×10^{-19}
K_{sp}	Barium iodate	$Ba(IO_3)_2$	6.0×10^{-10}

Molar masses needed: SrF_2 125.6 g/mol (17.78), HBr 80.91 g/mol (17.72).

End-of-Chapter Exercises

From the Textbook

Q. Which of these statements about the common-ion effect is most correct? (a) The solubility of a salt *MA* is decreased in a solution that already contains either M^+ or A^-. (b) Common ions alter the equilibrium constant for the reaction of an ionic solid with water. (c) The common-ion effect does not apply to unusual ions like SO_3^{2-}. (d) The solubility of a salt *MA* is affected equally by the addition of either A^- or a noncommon ion. ***[Exercise 17.58]***

Q. Use information from Appendix D to calculate the pH of (a) a solution that is 0.070 M in potassium propanoate (C_2H_5COOK) and 0.085 M in propanoic acid (C_2H_5COOH); (b) a solution that is 0.085 M in trimethylamine, $(CH_3)_3N$, and 0.10 M in trimethylammonium chloride, $(CH_3)_3NHCl$; (c) a solution made by mixing 25.0 mL of 0.20 M acetic acid and 25.0 mL of 0.25 M sodium acetate. ***[Exercise 17.59]***

Q. Which of the following solutions is a buffer? (a) 0.20 M formic acid (HCOOH), (b) 0.20 M formic acid and 0.20 M sodium formate (HCOONa), (c) 0.20 M nitric acid (HNO_3) and 0.20 M sodium nitrate ($NaNO_3$), (d) both b and c, (e) all of a, b and c. ***[Exercise 17.61]***

End-of-Chapter Exercises

From the Textbook

Q. (a) Calculate the pH of a buffer that is 0.150 M in lactic acid and 0.120 M in sodium lactate. (b) Calculate the pH of a buffer formed by mixing 75 mL of 0.150 M lactic acid with 25 mL of 0.120 M sodium lactate. ($K_a = 1.4 \times 10^{-4}$) ***[Exercise 17.62]***

Q. A buffer contains 0.20 mol of acetic acid and 0.25 mol of sodium acetate in 2.50 L. (a) What is the pH of this buffer? (b) What is the pH after the addition of 0.05 mol of NaOH? (c) What is the pH after the addition of 0.05 mol of HCl? ***[Exercise 17.65]***

Q. (a) What is the ratio of HCO_3^- to H_2CO_3 in blood of pH 7.4? (b) What is the ratio of HCO_3^- to H_2CO_3 in an exhausted marathon runner whose blood pH is 7.1? (K_{a1} of $H_2CO_3 = 4.3 \times 10^{-7}$) ***[Exercise 17.66]***

End-of-Chapter Exercises

From the Textbook

Q. Compare the titration of a strong, monoprotic acid with a strong base to that of a weak, monoprotic acid with a strong base (same initial concentrations). True or false? (a) More base is required to reach the equivalence point for the strong acid than the weak acid. (b) The pH at the beginning of the titration is lower for the weak acid than the strong acid. (c) The pH at the equivalence point is 7 no matter which acid is titrated. *[**Exercise 17.68**]*

Q. Predict whether the equivalence point of each of the following titrations is below, above or at pH 7: (a) $NaHCO_3$ titrated with NaOH, (b) NH_3 titrated with HCl, (c) KOH titrated with HBr. *[**Exercise 17.70**]*

Q. How many milliliters of 0.0750 M KOH are required to titrate each of the following solutions to the equivalence point: (a) 30.0 mL of 0.0900 M HCOOH, (b) 45.0 mL of 0.0750 M HNO_3, (c) 50.0 mL of a solution that contains 3.00 g of HBr per liter? *[**Exercise 17.72**]*

End-of-Chapter Exercises

From the Textbook

Q. A 10.0 mL sample of 0.250 M acetic acid is titrated with 0.100 M KOH. Calculate the pH after the following volumes of base have been added: (a) 0 mL, (b) 12.5 mL, (c) 24.5 mL, (d) 25.0 mL, (e) 25.5 mL, (f) 30.0 mL. ***[Exercise 17.74]***

Q. (a) If the molar solubility of CaF_2 at 35 °C is 1.24×10^{-3} mol/L, what is K_{sp} at this temperature? (b) It is found that 1.1×10^{-2} g SrF_2 dissolves per 100 mL of aqueous solution at 25 °C. Calculate the solubility product for SrF_2. (c) The K_{sp} of $Ba(IO_3)_2$ at 25 °C is 6.0×10^{-10}. What is its molar solubility? ***[Exercise 17.78]***

Q. Using Appendix D, calculate the molar solubility of AgI in (a) pure water, (b) 0.15 M KI solution, (c) 5.0×10^{-2} M $AgNO_3$ solution. ($K_{sp} = 8.3 \times 10^{-17}$) ***[Exercise 17.80]***

End-of-Chapter Exercises

From the Textbook

Q. (a) Will $Ca(OH)_2$ precipitate from solution if the pH of a 0.050 M solution of $CaCl_2$ is adjusted to 8.0? (b) Will Ag_2SO_4 precipitate when 100 mL of 0.050 M $AgNO_3$ is mixed with 10 mL of 5.0×10^{-2} M Na_2SO_4 solution? ***[Exercise 17.86]***

Q. A solution contains 1.0×10^{-4} M $Ca^{2+}(aq)$ and 1.0×10^{-4} M $La^{3+}(aq)$. If NaF is added, will CaF_2 ($K_{sp} = 3.9 \times 10^{-11}$) or LaF_3 ($K_{sp} = 2 \times 10^{-19}$) precipitate first? Specify the concentration of $F^-(aq)$ needed to begin precipitation. ***[Exercise 17.88]***

Q. A solution containing several metal ions is treated with dilute HCl; no precipitate forms. The pH is adjusted to about 1, and H_2S is bubbled through; again, no precipitate. The pH is then adjusted to about 8 and H_2S is bubbled through; this time a precipitate forms. The filtrate is treated with $(NH_4)_2HPO_4$; no precipitate forms. Which of these cations are possibly present or definitely absent: Al^{3+}, Na^+, Ag^+, Mg^{2+}? ***[Exercise 17.90]***